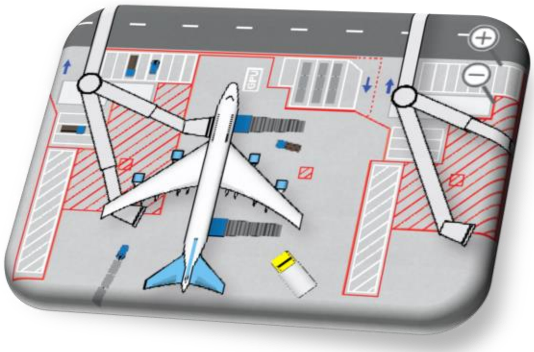

Airside Safety Course

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Description

The purpose of this course is to provide learners with an understanding of the concepts associated with Airside Safety.

Have 3 Parts of Course

Part 1 : The Airport Environment

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2. Communication
3. Hand Signals
4. Aircraft
5. Ground Support Equipment

Part 2 : Safety and Security

1. Security
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3. Health and Safety
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INTRODUCTION

Working at an airport means that you have to behave and operate according to many rules and regulations. In order to understand why these rules and regulations are important for you, it is necessary that you know a bit more about the 'Airport Environment'.

In this module you will be introduced to the different aspects of the airport and aircraft.

After completing this module you will be able to understand and communicate with other people working with aircraft.

The following slides are a pre-assessment to help you assess your own knowledge before taking the course. The questions will help prepare you for the material to be learned in this module. Your score is not recorded; it is purely for your own use for self-evaluation.

Begin the assessment.

Abbreviations

ID	Identification
UTC	Universal Time Co-ordination
GMT	Greenwich Mean Time
GSE	Ground Support Equipment
ULDs	Unit Load Dives
GPU	Ground Power Unit
MDL	Main Deck Loader
LHO	Live Human Organs
MTOW	Maximum Take-off Weight
MZFW	Maximum Zero Fuel Weight
MLAW	Maximum Landing Weight
DOW	Dry Operating Weight
TOF	Take-off Fuel
TTL	Total Traffic Load
EIC	Equipment in Compartment
LMC	Last Minute Change
LDM	Load Distribution Message
CPM	Container and Pallet Distribution
UCM	Unit Container Message
LIR	Loading Instruction
ACB	Anti-Collision Beacon
PPE	Personal Protection Equipment
HVC	High Visibility Clothing
FOD	Foreign Object Debris
ATC	Air Traffic Control
ERA	Equipment Restricted Area

DEFINITIONS

When the following terms are used in the Airside Safety Manual, they have the following meanings:

Accident is an undesirable incidental and unplanned event that could have been prevented had circumstances leading up to the accident been recognized, and acted upon, prior to its occurrence. Most scientists who study unintentional injury avoid using the term "accident" and focus on factors that increase risk of severe injury and that reduce injury incidence and severity

Aerodrome A defined area on land or water (including any buildings, installations and equipment) intended to be used either wholly or in part for the arrival, departure and surface movement of aircraft.

Airside is any customs controlled area such as the Departure and Arrival Hall and the area where the aircraft are loaded and serviced between arrival and departure.

Airplane (informally **plane**) is a powered, fixed-wing aircraft that is propelled forward by thrust from a jet engine or propeller.

Cargo is any property carried or to be carried in an aircraft. Air cargo comprises air freight, air express and airmail.

First Aid is the assistance given to any person suffering a sudden illness or injury, with care provided to preserve life.

Fire is the rapid oxidation of a material in the exothermic chemical process of combustion, releasing heat, light, and various reaction products. Slower oxidative processes like rusting or digestion are not included by this definition.

Health is the level of functional or metabolic efficiency of a living organism. In humans it is the ability of individuals or communities to adapt and self-manage when facing physical, mental or social challenges.

Incident An occurrence, other than an accident, associated with the operation of an aircraft which affects or could affect the safety of operation.

Hazard is a situation that poses a level of threat to life, health, property, or environment.

Landside Is the area of the airport terminals outside any Security Access Point such as Customs or Immigration Checkpoints.

Safety The state, in which risks associated with aviation activities, related to, or in direct support of the operation of aircraft, are reduced and controlled to an acceptable level.

Security means to control the access to property and aircraft and protect them from acts of vandalism, aggression and terrorism.

Weather is the state of the atmosphere, to the degree that it is hot or cold, wet or dry, calm or stormy, clear or cloudy.

QUESTION 0

the Landside is the area of the airport terminals ____________ any Security Access Point.

Select the best answer.

A) Inside
b) Outside
c) Within
D) Encompassing (X)

QUESTION 1

Using the 24 hour clock, how would you express 9 pm.

Select the best answer.

a) 1900
b) 0900
c) 2100
d) 2200

QUESTION 2

A marshaller is responsible for providing standard marshalling signals, in a clear and precise manner, to arriving and departing aircraft.

Select the best answer.

a) True
b) False

✍ QUESTION 3

Which component controls the aircraft movement around the horizontal longitudinal axis?

Select the best answer.

a) Ailerons
b) Elevators
c) Rudder
d) Flaps

✍ QUESTION 4

What type of GSE is used for pushback operations?

Select the best answer.

a) Jet starter
b) Aircraft tug
c) Passenger bridges
d) A mobile conveyer

Part 1 The Airport Environment

Section 1 The Airport

INTRODUCTION - THE AIRPORT (ICAO an 14)

- If you work at an airport there are a lot of rules and restrictions that you have to take into account.
- BE AWARE - some of these rules are of life and death importance!
- In this lesson you will be introduced to the basic airport terms.

LEARNING OBJECTIVES

After completing this lesson you will:

- Explain the difference between Landside and Airside
- Describe the different areas that comprise the Airside.
- Explain what the different colors and patterns of the airport markings mean.

✍ WHAT do you think?

✍ QUESTION 5

What do you think is true?

The area where aircraft land and take-off is called Landside.

a) The customs controlled area at an airport is called the Airside.
b) An airport is divided in two parts: Landside and Airside.

Correct. In this lesson you will learn about the different areas within the airside where the aircraft is handled. Read more on following screens.

1.1 LANDSIDE AND AIRSIDE

An airport is divided into two sections:

- Landside is the area of the airport terminals outside any Security Access Point such as Customs or Immigration Checkpoints.
- Airside is any customs controlled area such as the Departure and Arrival Hall and the area where the aircraft are loaded and serviced between arrival and departure.

In the following pages you will learn more about the various Airside areas.

1.2 AIRSIDE AREA

The Airside area is split into different sections:

- The Maneuvering Area
- The Apron
- Service roads

Find out in the following screens what and where these different sections are.

1.3 MANOEUVRING AREA

The ‘road’ that is used by aircraft to take off and land is called the Runway.

Under no circumstances are unauthorized vehicles allowed to drive on a Runway. Crossing Runways is only allowed under strict conditions.

Aircraft also use roads to taxi to and from a Runway or the Apron. This road is called the Taxiway.

The area used for

- take-off, landing and
- taxiing is called the
- Maneuvering Area.

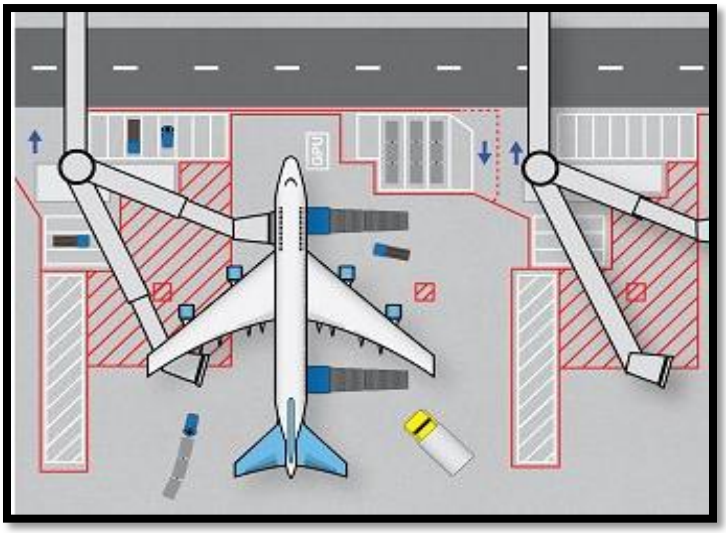
1.4 ENTERING THE MOVEMENT AREA

Vehicles are only allowed to enter the Maneuvering Area at controlled crossings. A double white line marks the boundaries of a Maneuvering Area. These crossings are controlled by use of radio or by traffic lights, If there are no traffic lights, vehicles must:

- Be equipped with a radio capable of receiving and transmitting on the appropriate Ground Frequency.
- Receive the necessary approval for entering from Air Traffic Control.

1.5 APRON AREA

The Apron is the area used for loading, unloading and parking aircraft. These ‘parking spots’ are usually called Gates, Bays, or Stands. The roads that are used by vehicles for servicing Aircraft are called Service Roads. These roads are not accessible to Aircraft. Aircraft can cross Service Roads at crossings controlled by traffic lights or stop bars. The combined area of Apron and Service Roads is called the Ramp.



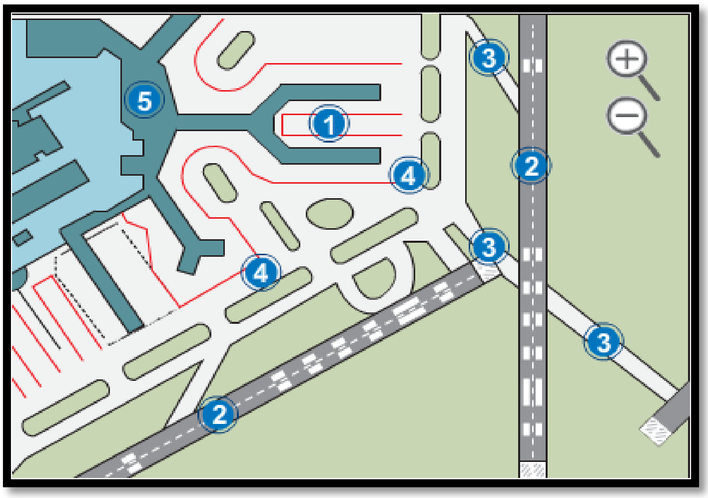
1.5.1 EMPLOYEE RESPONSIBILITIES

There are many people working many different jobs on the ramp. Click on a job title to find out more about it:

- **Aircraft Movement team:** in charge of guiding the aircraft to its position; in charge of the pushback.
- **Aircraft grooming:** a service offered by the ground handling company. They are in charge of cleaning the passenger area, galleys and toilets.
- **Flight dispatcher:** in charge of the flight plan, distributing the load and issuing the load sheet.
- **Airport Ramp Service Agents:** in charge of sorting the baggage into containers or baggage carts, in charge of loading and unloading aircrafts, driving the GSE.
- **Ramp Lead:** usually in charge of verifying that all procedures are done correctly during ground handling.
- The **Co-Pilot** or **First Officer** assists the Captain during the flight. He or the captain will check the outside of the aircraft before departure to be sure there is no visible damage to the fuselage.
- The **Captain** or **Commander** is responsible for the security of the whole flight, of verifying al the documentation and flying the passengers safely to their destination.

1.6 MOVEMENT AREA

The Apron and the Maneuvering Area combined is called the Movement Area.



NOW YOU KNOW!

Now you have learned about the different areas at the Airside of the airport.

Are you able to give the correct answers to the following questions?

QUESTION 6

An aircraft is loaded at the ____________.

a) Apron
b) Maneuvering area
c) Taxiways
d) Service roads
e) Movement area
f) Ramp

QUESTION 7

Aircraft cannot go on the ____________ .

a) Apron
b) Maneuvering area
c) Taxiways
d) Service roads
e) Movement area
f) Ramp

QUESTION 8

The routes that aircraft use to get to the runways are known as ____________ .

a) Apron

b) Maneuvering area

c) Taxiways

d) Service roads

e) Movement area

f) Ramp

✍ QUESTION 9

The Apron Area and the Service Roads are also known as the ____________ .

a) Apron

b) Maneuvering area

c) Taxiways

d) Service roads

e) Movement area

f) Ramp

1.7 MARKINGS AND BOUNDARIES AT AN AIRPORT (ICAO an 14 V. 1 & Aerodrome Design Manual)

Markings and boundaries may differ at individual airports. However, for all International Airports, the following markings are recommended:

- Yellow lines are used for guidance of the aircraft.
- White lines are used for vehicles. A double white line or an offset white line may not be crossed.
- Red lines are used to signify safety warnings.
- Solid red lines may not be crossed during aircraft movement. When required they should be crossed with caution.

For people working on the Ramp the following markings are important:

Double white line: Do not cross.

Single white line: Cross with caution.

Broken white line: A roadway centre-line.

Solid single red line: Do not cross during aircraft movement, cross with caution when required.

It is critical that you know these markings!

1.7.1 RED CLEARANCE LINE

The Red Clearance Line is a 60 cm wide uninterrupted red line. It marks the border between the Apron and the Maneuvering Area. This line should never be crossed by any vehicle other than aircraft or vehicles used for towing and pushback operations.

Remember: Crossing the Red Clearance Line means that you are entering the Taxiways and/or Runways.

NOW YOU KNOW!

QUESTION 10

The Clearance line marks the border between the Apron and the ____________ .

a) Service roads
b) Maneuvering area
c) Ramp
d) Movement area

QUESTION 11

The crossing points to enter the Maneuvering Area are marked by a ____________ .

a) 60 cm wide red line
b) Double white line
c) Single white line
d) Yellow line

Section 2 COMMUNICATIONS

INTRODUCTION – COMMUNICATION (ICAO Doc 9432 Manual of Radiotelephony)

The airport environment is an international environment where, because of all the different languages, confusion might easily occur.

To avoid confusion international standards and rules for communication have been introduced.

In this lesson you will learn how to communicate and avoid confusion.

LEARNING OBJECTIVES

After completing this lesson you will:

- Describe what the Phonetic Alphabet is and why it is used.
- Demonstrate how to express time and dates in aviation.
- Explain what time zones are and how to express time using the UTC standard.
- Explain what is meant by Zulu time.

WHAT DO YOU THINK?

QUESTION 12

To avoid confusion the Phonetic Alphabet was introduced.

What do you think the Phonetic Alphabet is?

a) A way to pronounce the letters of the alphabet.

b) A way to express the letters of the alphabet by making visual gestures.

1.2.1 PHONETIC ALPHABET

The Phonetic Alphabet is a way to pronounce the letters of the alphabet. It is useful when, for example, you have to spell a word by radio or phone. Just think how easily the letters 'M' and 'N' can be confused when spoken aloud.

1.2.2 THE PHONETIC ALPHABET PRONUNCIATION

All the letters of the Phonetic Alphabet are known by an associated name. The pronunciation of these names sounds the same everywhere, even when transmitted on a radio or pronounced with an accent or different dialect.

1.2.3 AVIATION TIME

When an aircraft flies between different time zones, major confusion with expressing the exact time of arrival can occur.

To avoid confusion, all operational timings in aviation are standardized.

- The 24-hour clock is used.
- All timings are expressed as Universal Time Co-ordination (UTC).
- Learn more about the expression of time and dates on the following screens.

1.2.4 24-HOUR CLOCK

All times in aviation are expressed using the 24-hour clock.

For example:

02.00 PM = 1400 hrs.

12 Midnight = 0000 hrs.

1.2.4.1 GMT: GREENWICH MEAN TIME

The world is divided into many different Time Zones.

The time in all these zones is related to the time at the 0- degree meridian, which runs through the town of Greenwich (10 km east of London, UK).

The time at the 0-degree meridian has traditionally been referred to as Greenwich Mean Time (GMT). It is also sometimes referred to as Greenwich Meridian Time.

1.2.4.2 UTC: UNIVERSAL TIME CO-ORDINATION

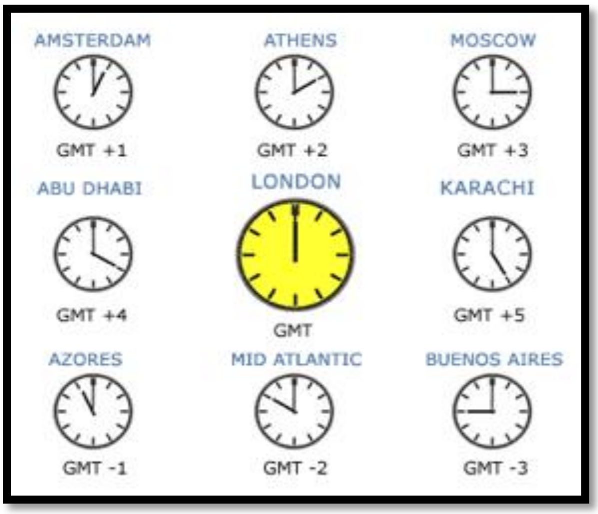
The current, internationally accepted method of expressing the exact time at the 0-degree meridian is Universal Time Co- ordination (UTC).

Before the introduction of UTC as the standard expression for time, GMT was used. GMT and UTC are exactly the same; however GMT is not officially used anymore.

One time zone east of Greenwich will be indicated as UTC +1, and means that it is one hour later in that zone.

One time zone west of Greenwich will be indicated as UTC –1 and means that it is one hour earlier in that zone.

The UTC is used throughout the world and does not change for daylight saving in summer.



1.2.4.3 ZULU TIME

In operational messages, UTC/GMT can also be expressed with the suffix Z. The letter "Z" (phonetically "Zulu") refers to the time at the prime meridian (0-meridian).

For example: 0900 UTC= 0900 Z

The above time can be expressed as:

- 0900 *ZED*
- 0900 *Zulu*
- 0900 *UTC*

1.2.5 TIME & DATES

A standard expression for the time and date is also used. To express the date, the day of the month is always put before the time.

When a flight departs at 2300 on Monday April 10 and arrives at 0030 on Tuesday April 11 this will be expressed as:

dep. 102300 - arr. 110030

SUMMARY

In this lesson you have learned how to communicate about times and dates and how to use the phonetic alphabet:

- When spelling a word aloud, you use the Phonetic Alphabet.
- When communicating about time and dates, you use the 24-hour clock and the UTC time.
- When using the date, you put the day of the month before the time.

Section 3 HAND SIGNALS

INTRODUCTION - HAND SIGNALS (ICAO an 2 / IATA AHM 631)

We have all seen them! Those people standing in front of an airplane, making gestures to guide the aircraft into a stand.

Who is allowed to do this and what do these hand signals mean?

LEARNING OBJECTIVES

After completing this lesson, you will:

- Explain what hand signals are used for and who uses them.
- Explain what Marshalling Hand Signals are and how they are used.
  - List the Marshalling Hand Signals.
- Explain what Servicing Hand Signals are and who uses them.
  - List the Servicing Hand Signals.
- Describe the hand signals used during Pushback Operation.

✍WHAT DO YOU THINK?

✍ QUESTION 13

Which of the following statements is correct?

a) International Orange' wands, table tennis bats, or gloves MUST be used for ALL signaling by ALL participating ground staff to aircraft.

b) Hand signals are only allowed for guiding aircraft when departing.

c) All ground staff use hand signals to communicate to aircraft.

1.3.1 HAND SIGNALS

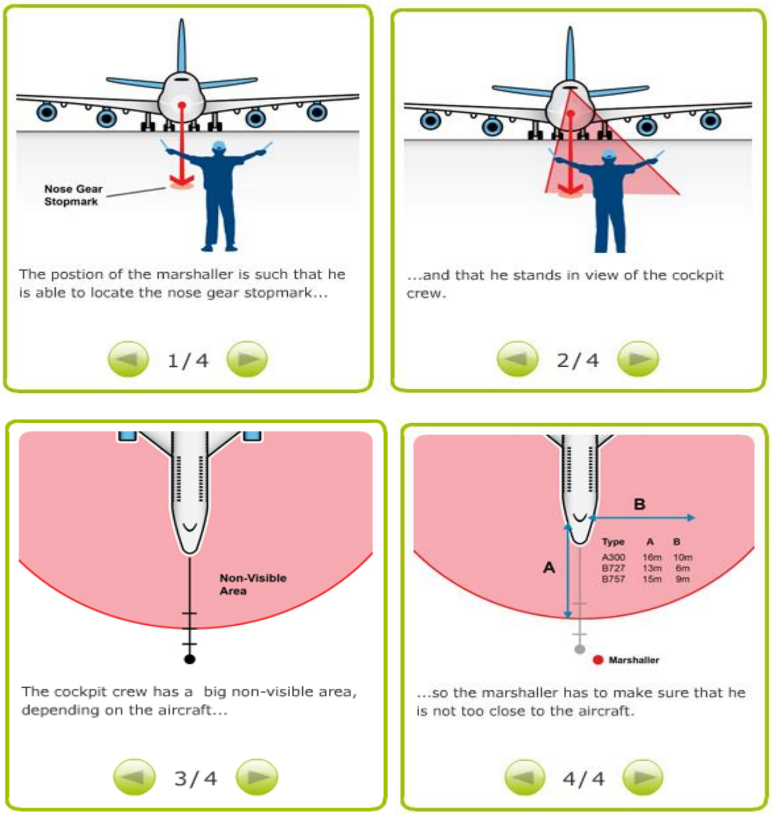
Hand signals are used for communication between the ground staff and the cockpit crew. These hand signals may only be performed by qualified and trained persons.

There are different types of hand signals:

- Aircraft marshalling.
- Technical/servicing signals.
- Ground crew pushback communication.
- Be aware that hand signals will also be used for communication during vehicle maneuvering.

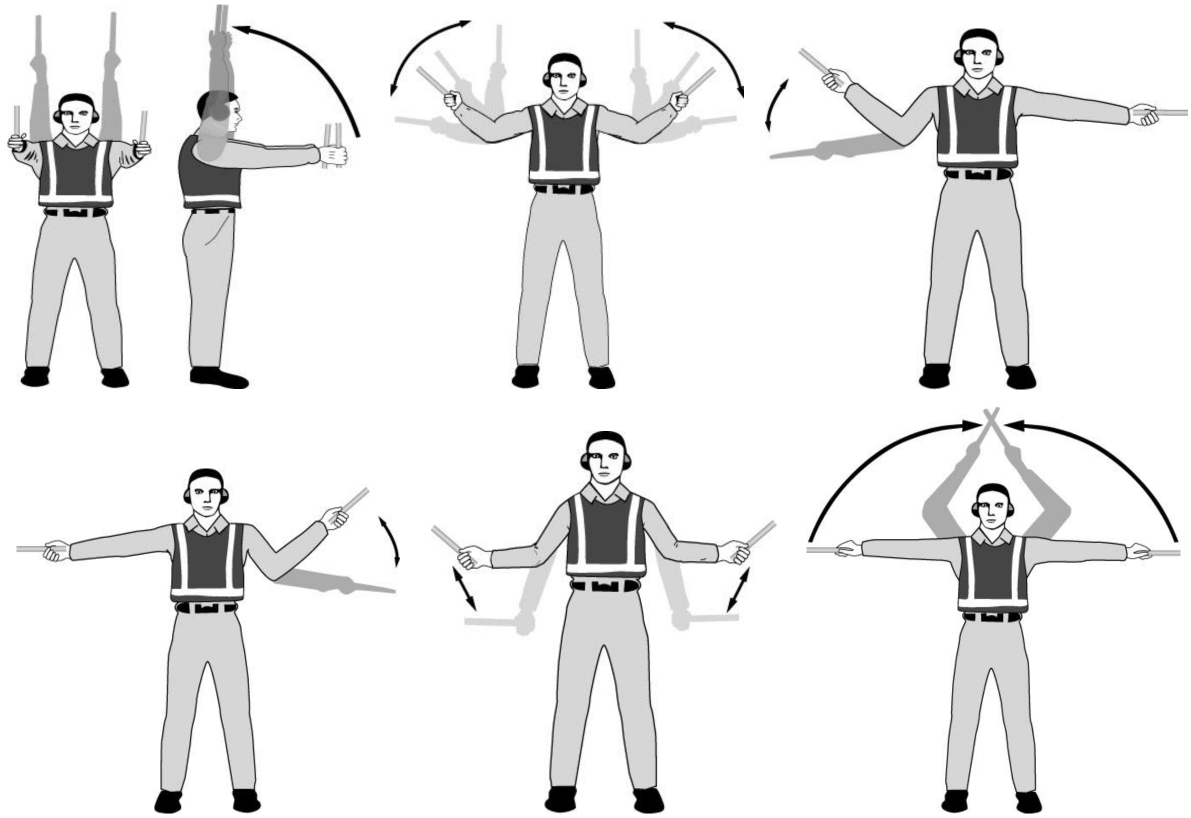
1.3.2 LIMITED VIEW

Marshalling hand signals are used to help the pilot maneuver the aircraft into the stand. This is necessary because the pilot's vision is limited directly around the aircraft. Be aware that there is limited visibility from the flight deck - the Marshaller will be standing some distance away from the aircraft.



1.3.3 MARSHALLING

No person should attempt to marshal or guide an aircraft unless trained, qualified, and approved to carry out such functions. Marshallers should be authorized to perform their tasks by the appropriate Airport Authority. A marshaller will be responsible for providing standard marshalling signals, in a clear and precise manner, to arriving and departing aircraft.



1.3.4 WING WALKER

During arrival no staff or vehicle may pass between the marshaller and the aircraft being marshaled.

Before the aircraft enters the designated gate area, the wing walkers must confirm that it is clear for the aircraft to move using the correct hand signal.

During arrival, if the marshaller has any doubts about obstructions in the aircraft path, he should check with the wing walkers and stop the aircraft movement until any doubt has been cleared.

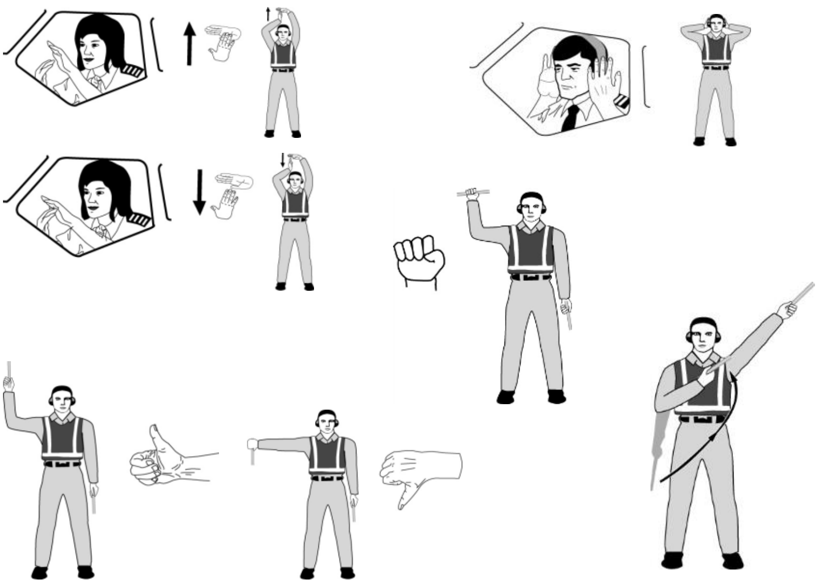
1.3.5 SERVICING HAND SIGNALS

Not all hand signals between ground staff and cockpit are marshalling hand signals. Additionally, during (technical) servicing of the aircraft, the use of hand signals is a very important way of communicating. These signals are called technical or servicing hand signals. These hand signals should only be used when verbal communication is not possible.

1.3.6 TYPES OF SERVICING HAND SIGNALS

These servicing hand signals indicate the following to the flight crew:

- (Dis.)connect Ground Power
- Affirmative (all clear)
- Negative (all clear)
- Interphones.
- Do not touch controls.
- Open/close stairs forward/aft.
- These hand signals are based on IATA standards and might have local variations.



1.3.7 COMMUNICATION DURING PUSHBACK

- Hand signals are also used for communication during pushback operations. A pushback operation is the reversing of an aircraft that is loaded with cargo or passengers from the Parking Position to the Taxi Position. The pushback will be done with a pushback tug.
- During pushback no vehicle may pass behind the aircraft or through the operational safety zone.

1.3.8 PUSHBACK OPERATION

Before pushback of the aircraft is initiated, the wing walkers must confirm to the headset operator that it is clear for the aircraft to move using the correct hand signal. The operational safety zone must be free of all obstacles.

A pushback operation can be performed by 2 persons:

  - A headset operator
  - A tug driver

- However, in some cases, a single operator pushback may be approved. In this situation the operator will use a headset. The headset operator overlooks the situation on the ground and maintains contact with the flight deck using a headset.
- Remember: during departure, if the headset operator has any doubts about obstructions in the aircraft path, he should check with the wing walkers and stop the aircraft movement until any doubt has been cleared.

1.3.9 WING WALKER DUTIES

- At some airports it is necessary to have two Wing walkers during pushback. They are responsible for:
- Watching for any danger to the aircraft during pushback (vehicle traffic, GSE, staff, etc.).
- Ensuring and maintaining the safety clearance of the aircraft during movement and advise the headset operator to stop the
- Aircraft in case of danger.

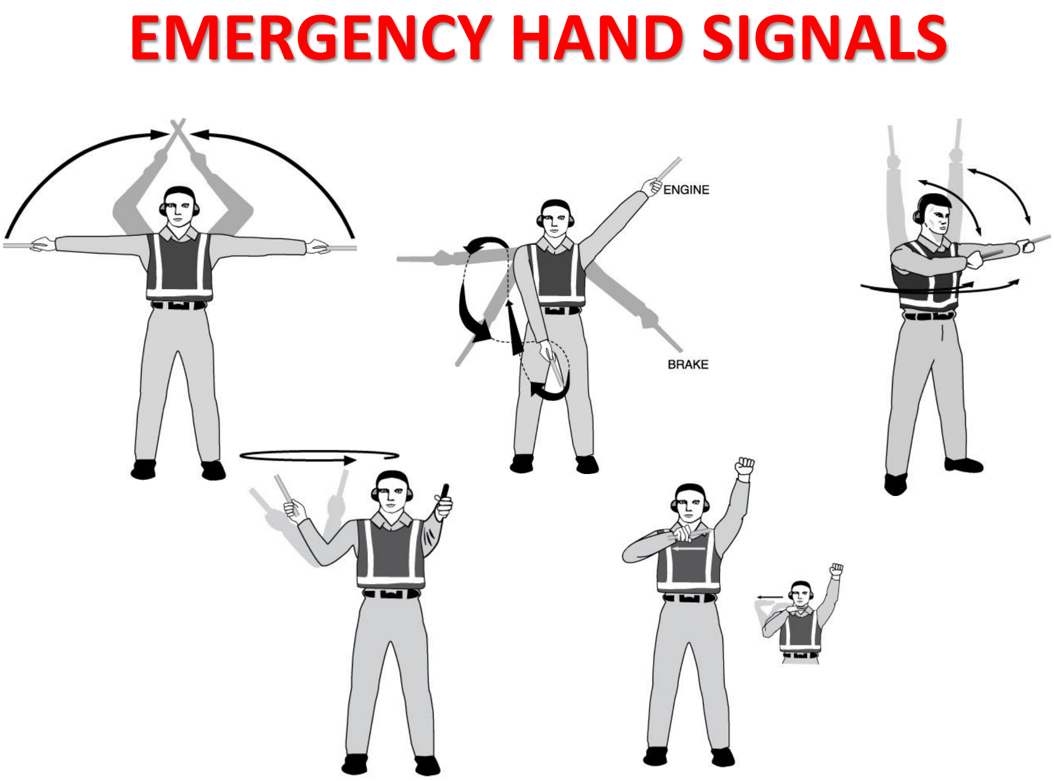
1.3.10 END OF PUSHBACK

- Many airlines require a final marshaller. Once the aircraft has arrived to its final position after pushback, the marshaller will hold the stop hand signal until all the equipment and the area is clear of hazards and obstructions.
- Then the marshaller will change his position to a hold /stand by position until the crew indicates they are ready to taxi.
- After checking that the area is clear, the marshaller changes to the end marshalling sign.

End marshalling: Perform a standard military salute with right hand and/or wand to dispatch the aircraft. Maintain eye contact with the flight crew until the aircraft has begun to taxi.

Hold position/Standby: Fully extend arms and wands downwards at a 45° angle to the sides. Hold the position until the aircraft is clear for the next maneuver.

1.3.11 EMERGENCY HAND SIGNALS



1.3.12 NOW YOU KNOW!

QUESTION 14

Which of the following statements is correct?

a) Servicing hand signals are only used when verbal communication with the cockpit is not possible.

b) The headset operator uses hand signals for communicating with the flight crew and the tug driver.

c) The hand signals that the wing walker makes are only to give instructions about the movement of the aircraft.

1.3.13 ALWAYS TAKE CARE

When a marshaller, headset operator, or wing walker is concentrating on guiding the aircraft they will not be able to watch out for other vehicles.

It is therefore very important to always give way to persons involved in hand signaling. Never drive between the marshaller and the aircraft or obstruct the vision of the marshaller or the pilot.

Take good notice of the direction in which the aircraft is being guided.

SUMMARY

There are different types of hand signals:

- Aircraft marshalling: These signals are used by a marshaller to guide the aircraft into and out of the stand when it arrives and departs.
- Technical/servicing signals: These signals are used to communicate with the cockpit during aircraft servicing.
- Ground crew pushback communication: These signals are used by personnel involved in pushback operations.
- When you see an aircraft being guided by hand signals you should stay clear of the aircraft as well as the person signaling and make sure not to obstruct their vision in any way.

Section 4 AIRCRAFT

INTRODUCTION – AIRCRAFT (IATA AHM 400)

Probably the most impressive 'object' you can come across at the airside is the Aircraft. In this lesson you will be introduced to basic knowledge about Aircraft. This lesson is about the relevant aircraft terms and features, which you need to, know when working safely at an airport.

LEARNING OBJECTIVES

After completing this lesson you will:

- Define the different positions and parts of an aircraft and explain how they are named.
- Describe the different classes of aircraft and what distinguishes them.
- List the different sections of the fuselage and explain what they are used for.
- Describe the different doors and hatches and their location and function on an aircraft.
- List the critical aircraft parts and their location on the aircraft.
- Describe the primary control surfaces that control the aircrafts movement and explain how they control the aircraft's movement.
- Describe the components and locations of the landing gear.

WHAT SIDE ARE YOU ON?

Did you know?

The indication of the positions left or right of an aircraft is always taken from the perspective of the cockpit (looking ahead).

- Left of the aircraft is called: Port.
- Right of the aircraft is called: Starboard.
- The body and the top of an aircraft is called: Fuselage.
- The part of the fuselage in front of the wings is called: Forward.
- The part of the fuselage behind the wings is called: Aft.

1.4.1 DIFFERENT CLASSES

Aircrafts are divided into two classes:

- Wide-bodied aircraft.
- Narrow-bodied aircraft.

Apart from the size, the primary distinction between the two classes of aircraft is the way the doors are operated:

- Narrow-bodied aircraft have manually operated doors.
- Wide-bodied aircraft have hydraulically operated doors.

- Another distinction is that **wide-bodied** aircrafts have <u>two aisles</u> in the passenger section and **narrow-bodied** aircrafts have a <u>single aisle</u>.
- The class to which an aircraft belongs to also impacts the procedures and equipment used for servicing the aircraft.

1.4.2 WIDE-BODIED AIRCRAFT

The following aircraft are classified as Wide-bodied:

- Airbus: A300 / A310 / A330 / A340 / A380.
- Boeing: 747 Jumbo Jet / 767 / 777.
- McDonnell Douglas (Boeing): DC-10 / MD-11.
- Lockheed: L-1011 TriStar.
- Ilyushin: IL-86 / IL-96
- All other aircraft can be classified as Narrow-bodied.

Airbus: A310

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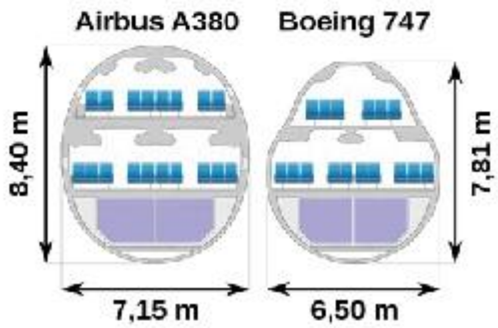

Airbus A380 and Boeing 747 comparision

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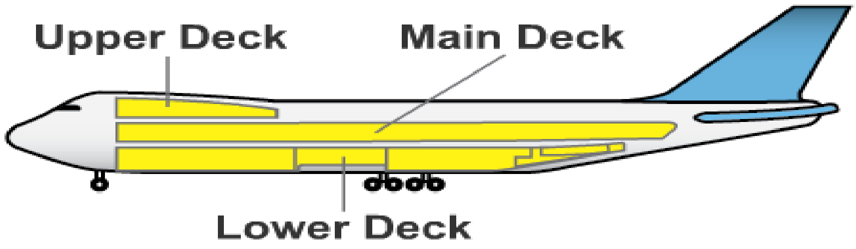
1.4.3 FUSELAGE

The fuselage (or body) of the aircraft contains three sections:

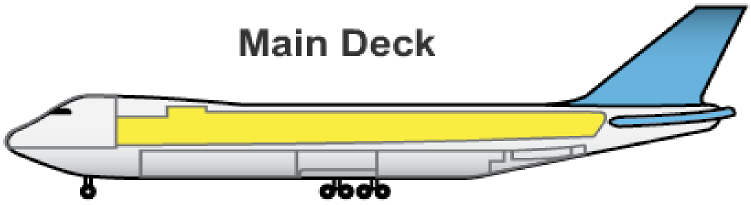
- The main deck.
- The lower deck.
- The upper deck.



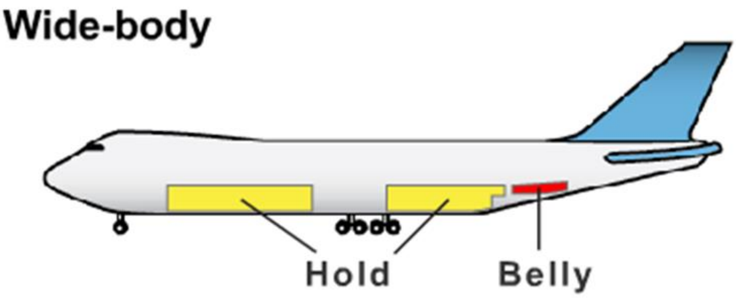
1.4.4 MAIN DECK

In a passenger aircraft the main deck and the upper deck are used as cabin area, where the passengers are seated.

In a cargo aircraft the main deck is used for carrying cargo, pallets and containers



1.4.5 LOWER DECK AND BELLIES



1.4.6 UNIT LOAD DEVICES

The pallet and container units that are used to stow baggage, cargo and mail in the aircraft are called Unit Load Devices (ULDs).

Because there are many different types of aircraft that carry cargo, as well as different possible configurations within the same aircraft type, most ULDs are specific to a particular use.

Proper handling of a ULD.

ULD damage caused by improper handling

1.4.7 PASSENGER AND SERVICE DOORS

An aircraft has different doors. To avoid confusion it is important to know the names and the locations of these doors. The most significant distinction is made between passenger doors and service doors:

- The passenger doors are normally found on the left side of the aircraft (the port side).

- The number of passenger doors varies depending on the size and type of the aircraft. The service doors are normally located directly opposite the passenger doors, on the starboard side. These doors are only used for services, such as catering and cleaning.

1.4.8 CARGO AND BELLY DOORS

Cargo is placed into an aircraft through a cargo door or a belly door. But there are some major differences between a cargo door and a belly door. Cargo doors are mechanically operated doors that give access to the main deck loading area or bulk holds. These doors may only be operated by specially trained staff or the aircraft crew.

Belly doors are always manually operated and give access to the lower deck bulk cargo area. Belly doors are mostly found on Narrow-bodied aircraft. The rear, “bulk” hold or compartment door is similar to narrow-bodied aircraft.

1.4.9 CRITICAL AIRCRAFT PARTS

For your own safety it is of vital importance to be very careful around aircraft, particularly the engines. When an aircraft is at an airport for servicing, handling, or loading, there are also other aircraft parts around which it is very important to be careful. This is for your own safety and for the safety of crew and passengers during the flight. Even the smallest damage could endanger the safety of the aircraft and those on board during the flight.

In the next pages you will learn about these critical aircraft parts, where they are located, what their function is, and how you can avoid damaging them. Take good notice, it is of great importance for flight safety.

1.4.10 WINGS AND STABILISERS

The wings and stabilizers are the main parts that enable the aircraft to take off and fly. The wings and stabilizers have primary and secondary control surfaces.

The three primary control surfaces that control the aircraft's movements around the three main axes are:

- Ailerons, for movements around the horizontal longitudinal axis.
- Elevators, for movements around the horizontal cross axis.
- Rudder, for movements around the vertical axis.

1.4.11 AUXILIARY SURFACES

In the previous animation you have seen how the ‘primary control surfaces’ are relevant for the movement of the aircraft. To support these movements there are also ‘helper’ parts in the wings, called spoilers and flaps. In particular the ‘moveable’ flaps need special caution when you are working at an aircraft.

1.4.12 FLAPS

Flaps extend downwards from the back of the wing. When they are deployed, clearance can be significantly reduced. Always use extreme caution in the area where the flaps can be extended, as damage to any part of the aircraft can prevent it from flying.

1.4.13 LANDING GEAR

In this last section of this technical lesson you will learn about the landing gear. The landing gear consists of struts with wheels attached under each wing and the nose wheel. The main landing gear is the part of the aircraft that touches the runway first during landing. The main landing gear is designed to take large impacts (and heavy forces).

The nose gear is the component that is designed to steer the aircraft on the ground. It is not designed to take large loads and impacts.

Each unit retracts into the wing and fuselage. On Wide-bodied aircraft there may be additional sets of wheels directly underneath the fuselage.

SUMMARY

In this lesson you have learned about the different parts on the aircraft, including:

- The 4 sides of the aircraft and fuselage.
- The main deck, lower deck and flight deck.
- Wide-bodied and Narrow-bodied aircraft.
- Passenger, service, cargo, and belly doors.

And you have some basic knowledge about different parts of the aircraft, including:

- Wings and stabilizers.
- Flaps.
- Landing gear.

Section 5 GROUND SUPPORT EQUIPMENT

INTRODUCTION - GROUND SUPPORT EQUIPMENT (IATA AHM 630)

As you know by now, airports can be a hazardous working place if you are not careful! In this lesson you will be introduced to the most relevant Ground Support Equipment that is used during an Aircraft Turnaround. You will learn about their function, and the possible dangers that you must be aware of.

LEARNING OBJECTIVES

After completing this lesson you will:

- Identify the most frequently used Ground Support Equipment (GSE).
- List the hazards and the danger zones of the GSE.
- Define and describe the following equipment:
  - Ground Power Unit
  - Catering truck, water truck, lavatory truck
  - Transporter and Dolly Train
  - Aircraft loading equipment
  - Refueling equipment
  - Bridges and stairs
  - Aircraft Tugs
  - Jet Starter
- Explain what safety precautions should be taken on the ramp around Ground Support Equipment.
- Define the fueling zone and explain what precautions need to be taken there.

WHAT DO YOU THINK?

As you can see in this picture, a lot of GSE are involved during the servicing and turnaround of an Aircraft.

Where do you think the following GSE is positioned?

- Tow tug
- Fuelling
- Air conditioning
- Electricity (GPU)
- Transporter loader

1.5.1 MAIN HAZARDS

There is a lot of activity with many different moveables Ground Support Equipment during the aircraft turnaround.

The main hazards are:

- During Manual Handling (e.g. handling cargo & baggage)
- Falls from heights.
- Moving vehicles driving up to and away from aircraft.
- Fire and explosion when refueling of aircraft.
- Noise, Jet starter (Air starter), GPU and vehicle engines.
- Electricity, Ground power units and cables.
- Badly stowed cables, spillages of fluids (e.g. fuel, oil, hydraulic fuel)
- Moving parts of machinery (e.g. conveyor belts)
- Always take good care and be very aware of your own safety and that of others.

1.5.2 GROUND POWER UNIT (GPU)

A Ground Power Unit (GPU) provides electrical power when the aircraft engines are not running. During servicing of an aircraft this power is necessary for lightning, communications and operating the doors.

A GPU is a moveable power supply unit and it is wheeled or driven to its position at starboard near the cockpit. Heavy cables are plugged into the aircraft to supply the electricity. Be very careful with the (loose) Power cables and the plugs.

NOTE: a GPU generates high voltage electric power. Do not under any circumstances touch any electrical components that may contain current. This could be lethal.

REMEMBER: during the arrival of the aircraft the GPU equipment and cables must be outside the operational safety zone.

1.5.3 CATERING AND 'AQUA SERVICES

As mentioned earlier, a main hazard at the Apron is the danger of moving vehicles driving up to and away from the aircraft.

Not so surprising after all

Just imagine all the different trucks and equipment that are needed for loading:

- Cargo & mail
- Luggage
- Water
- Catering
- Fuel
- And, last but not least, passengers

1.5.4TRANSPORTER AND DOLLY TRAIN

The ground support equipment used for transporting cargo to and from the plane depends on the type of cargo (i.e.: containers and pallets (ULDs), loose cargo or baggage). The following equipment can be used.

✏ **Do Not's**

Always remember the following when working with:

Transporters:

a) Never walk on the rollers or castors of a transporter.
b) Never drive under the wing of an aircraft

Dolly or baggage cart trains:

Trains of dollies or baggage carts tend to 'drift in' or shorten turning corners. Stay clear of trains while they position or pass by.

- Never pass between dollies or carts While attached to a tractor, because it may Move unexpectedly and cause serious injury.
- Trains of dollies or baggage carts should never Pass under the wings of an aircraft

1.5.5 LOADING THE AIRCRAFT

To get the cargo and the baggage into the aircraft the following equipment can be used: A mobile (belt) conveyor is a self-propelled vehicle with a continuous belt. It is used to load cargo, mail and luggage in the bulk belly of the aircraft. The mobile conveyor can be raised at both ends.

A hi-lift elevator (or high loader) is a self- propelled mobile elevator. It is used to load containers and pallets on the aircraft.

A larger variant of the high loader is the so called main deck loader (MDL), which is used to load cargo into the main deck.

1.5.6 TYPES OF CARGO

In the world of cargo you will find that cargo will be sent either as bulk cargo or built up in a ULD. The types of cargo vary, you will have:

- General cargo and
- Special cargo
- **Special cargo**: is divided into dangerous goods and other special cargo. Under other special cargo you can find for example:
- **Live Animals:** The IATA Live Animal Regulations describe how live animals must be transported.
- **Valuable and vulnerable cargo:** due to their value you will have additional security procedures
- **Perishable cargo:** cargo that is sensitive to temperature and time. It may decrease if exposed to extreme changes in temperature or humidity or delayed in transport.
- **Human remains:** be careful to load according to the policies of your company

- Shipments of **special importance or urgency** like LHO (live human organs)
- **<u>Wet cargo:</u>** this is a term given to all load which contains liquids or from which liquids may come out because of the nature of the load (example frozen meat, live animals).

Remember when preparing the built up cargo you should minimize the possibilities of damage, pilferage and mishandling of all cargo.

<u>1.5.7 MAXIMUM WEIGHTS</u>

Aircraft have three maximum weights which must be taken into consideration when loading an aircraft. These weights protect the structure of the aircraft during three critical flight phases: take-off, flight, and landing.

The three weights are:

- Maximum Take-Off Weight (MTOW)
- Maximum Zero Fuel Weight (MZFW)
- Maximum Landing Weight (MLAW)

<u>1.5.8 MAXIMUM WEIGHTS COMPONENTS</u>

The Operational Take-Off Weight is calculated for every flight and includes the following components:

Dry Operating Weight (DOW): The weight of the aircraft plus standard crew and their baggage, flight equipment, water and the pantry.

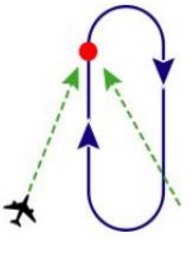
Take-Off Fuel (TOF): Includes the fuel on board at take-off which includes:

- Trip Fuel (fuel consumed between origin and destination)
- Contingency Fuel (used for unexpected bad weather conditions or other unexpected conditions)
- Alternate Fuel (used when you cannot land at your intended destination and must fly to an alternate destination)
- Holding Fuel (used when you cannot land immediately at destination, so the aircraft is in holding)
- Extra fuel: this fuel can be requested by the captain

Total Traffic Load (TTL): The weight of the passengers and their carry-on and checked baggage, cargo, mail and Equipment in Compartment (EIC)

<u>1.5.9 HOLDING</u>

In aviation holding (or flying a hold) is a maneuver designed to delay an aircraft already in flight while keeping it within a specified airspace.



A standard holding pattern, Shown are the entry (green), the holding fix (red) and the holding pattern itself (blue)

1.5.10 REFUELLING

Aircraft fuelling is a critical and very important activity during the aircraft turnaround period. The fuel tanks are positioned in the wings and in the centre part of the aircraft. Different GSE can be used for (re-)fuelling an aircraft. This includes:

A hydrant dispenser. This equipment is linked to the fuel distribution pipeline that is part of a hydraulic underground fuel farm. This equipment can fuel large amounts of kerosene within a very short time. The vehicles cannot be moved while connected to the aircraft or the underground pipeline.

A tank re-fueler. This is a tank truck that brings the kerosene to an aircraft. This will be used for smaller aircraft and if a hydraulic fuel supply system is not available. They are also used when defueling aircraft.

Once the aircraft has arrived and stopped at the nose gear stop mark and all arrival tasks have taken place, you will most probably start fuelling.

Remember: aircraft ground handling activities take place at the same time as aircraft fuelling operations. These activities must be compatible to ensure the safety and integrity of the operation.

The amount of fuel required is indicated in the flight plan and sometimes the captain will want to take some extra fuel.

- The fuel truck driver will be informed of the amount of block fuel (take off fuel + taxi fuel)
- Next the fuel truck driver will connect the fuel hose to the aircraft.
- The fuelling safety zone shall be regarded as an area extending 3 m (10 ft.) radially from the aircraft fuelling point, venting point, and fuelling equipment.
- The aircraft and the fuelling vehicles must be electrically bonded together throughout the fuelling operation.
- The hydrant pit valve shall be identified by a four winged flag or equivalent and clearly visible.
- The fuel operator must maintain control of fuelling operations using the hand held deadman device throughout the operation, remaining outside the vehicle cab at all times.
- Portable electronic devices, such as mobile telephones, portable radios and pagers, should not be used within the fuel safety zone.
- Remember to follow the procedures of the Airline in case of fuelling with passengers on board.

1.5.11 FUELING ZONE

The Fuelling Safety Zone is a 3 meter radius extending from the aircraft fuelling point, venting point, and fuelling equipment. In the Fuelling Safety Zone, smoking, matches, lighters, or other flammable material must not be on anyone's person when fuelling aircraft.

Equipment with ‘all metal wheels’ or producing sparks shall not be moved in the Fuelling Safety Zone while fuelling is in progress.

Make yourself fully aware of the emergency 'cut-off' switches and their location.

After fuelling, the weight of the wings has increased. This can reduce the clearance height of the wings, easily leading to damage to aircraft or equipment if care is not taken.

1.5.12 DEAD MAN'S SWITCH

A dead man's switch that is automatically operated in case the human operator becomes incapacitated, such as through death or loss of consciousness.

The switch, a form of fail-safe, is usually wired so that it stops a machine by breaking a series circuit; Switches are used in aircraft refueling.

1.5.13 BRIDGES AND STAIR

When the aircraft is almost ready for departure, the passengers can enter the aircraft. To enable the passengers and crew to enter, passenger bridges or mobile stairs are used.

- Apron passenger bridges are telescopic bridges between the terminal and the aircraft.
- Mobile stairs can be self propelled or towed by a truck. And are positioned adjacent to an aircraft.

Stairs and steps can also be used during handling activities.

Be aware: Never move stairs until ordered to do so by the crew. Never move stairs with people still standing on it. Before any action is undertaken, ensure that everyone involved is aware of what is happening.

If stairs are removed from an open door, always ensure the safety lanyard is in place.

1.5.14 AIRCRAFT TUGS

- The aircraft is about to leave the Apron for take-off. Usually the aircraft has to be 'pushed back' first. Pushback is the reverse movement of aircraft (with passengers and/or cargo load) from the parking position to taxi position.
- An aircraft tug is used for pushback and towing operations. There are many different types of tugs, varying in size, weight and function.

The different types are:

- The conventional aircraft tug that uses a towbar to attach to the aircraft. Never step over towbars connected to an aircraft nose gear and/or or a towing tug.
- Towbarless tugs attach to the aircraft by lifting the nose-gear off the ground.

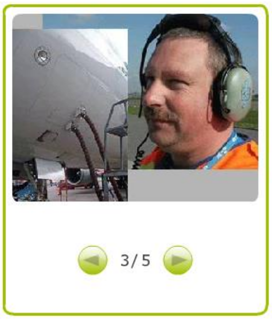
- Remember: if you are the wing walker, always be in sight of the tug driver and stay at least 3 meters away from the wings and fuselage of the aircraft.

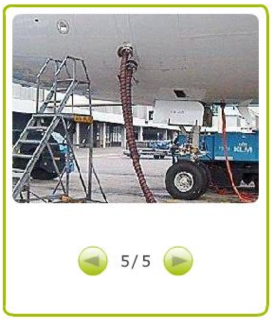
1.5.15 JET STARTER

A jet starter is used to support the starting of the aircraft engines. The jet starter provides the air that is required to start the engines.

A jet starter produces a lot of noise. Ear protection is required when working with a jet starter.

Make sure that the tubes of the jet starter are not bent or twisted before the air delivery commences. This can cause a powerful whiplash of the tubes that might result in personal injuries.





1.5.16 MESSAGES FROM OPERATIONS

Messages sent after the flight departs:

These messages are sent from the departure station to the next destination station. The messages are sent automatically after the flight dispatcher has corrected any LMCs (last minute changes) in the system. In case of system failure they must be sent manually.

1.5.16.1 LDM (Load Distribution Message):

The header contains flight number, departure date, registration, total seating capacity, origin and destination and number of crew members.

The load information is per destination and you will find the total amount of passengers divided by gender, the total dead load and then dead load per compartment. You will then find the total passengers per class including PADs and XCR if checked in. You will then find the total weight of all the baggage, cargo, mail and EIC as well as supplementary information about special load and rush bags.

1.5.16.2 CPM (Container and Pallet Distribution Message):

Message containing information about dead load on board per position and ULD type. It is used for all ULD aircraft single and multi sector flights. It serves as disposition helps at destination airport concerning loading staff and equipment. You will find all the loading positions, the type of ULD loaded, destination, weight and type of load (B, C, M, and E). You will also find the special load code if special load is loaded.

1.5.16.3 UCM (Unit Container Message):

Message containing information about ULDs with their ID- numbers that are being sent in that flight. It serves as a ULD stock control and is sent for every flight. This message only informs the ULD ID on board.

1.5.16.4 LIR (Loading Instruction):

This is issued for every flight before you can start loading. In this document you will find the positions, what will be loaded in each position (B, C, M, E), special remarks (PER, PEF, RFL, etc), weights, ULD ID in case of ULD aircraft.

1.5.17 ASSESSMENT - AIRPORT ENVIRONMENT

You have now finished the Airport Environment module.

In this module you have learnt about:

- Airport Environment
- Communication
- Hand Signals
- Aircraft
- Ground Support Equipment
- Passing this assessment will confirm completion of the Airport Environment Module and then you may continue to the next module.
- Keep in mind there is a FINAL EXAM at the end of this course following successful completion of all three (3) modules. The final exam will cover the material in greater detail.

Part 2 Safety & Security

COURSE OVERVIEW (ICAO an 14, 17 & ACI Airside safety)

Welcome to IATA's Basic Airside Safety course. This module, Safety and Security, is the second of three (3) modules you must complete to receive a certificate of completion.

This section of the module will introduce you to the features used throughout the course and how to navigate your way through the interface.

If you are already comfortable using the course, you may continue directly to a section by selecting it from the list below.

- Introduction
- Security
- Aircraft Danger Zones
- Health and Safety
- Severe Weather Conditions
- Traffic Rules at the Ramp
- Accidents and Incidents Reporting

INTRODUCTION

To move around on the airside of an airport is not without danger!

In this module you will learn more about the dangers and the Safety & Security Issues at an Airside. You are not allowed on a ramp if you have not passed the test at the end of this course.

PRE-ASSESSMENT

The following slides are a pre-assessment to help you assess your own knowledge before taking the course. The questions will help prepare you for the material to be learned in this module. Your score is not recorded; it is purely for your own use for self-evaluation.

QUESTION 15

If you see someone you don't recognize at the ramp, what should you do?

a) Nothing – there are so many people at the ramp you can't know them all.
b) Let your supervisor know that you saw someone you didn't recognize when you finish your shift
c) Ask the person for some identification to prove they can be on the ramp.
d) Ask your co-worker if they know them.

QUESTION 16

To produce the power that will create thrust with a jet engine, air is drawn into the front intake of the engine. This creates a huge suction which can be powerful enough to suck a human body into the engine.

- True
- False

QUESTION 17

When would you need to wear face protection when working on the ramp?

a) When you are working with gas
b) When you are working with dangerous fluids
c) When it's cold
d) You must always wear face protection

QUESTION 18

A lightning alert is issued when lightning activity is detected at a distance of 8 km. What should be done during this phase?

a) Continue normal operations
b) Stop fueling the aircraft
c) Suspend non-essential activities in open area
d) Suspend all activities on the ramp

QUESTION 19

An Airside Driving Permit is the same permit that allows you to drive on all roads.

- True
- False

QUESTION 20

An incident is an event that:

a) Results in death
b) Results in a disruption to safety
c) Results in injury
d) Results in damage

Section 1 SECURITY

INTRODUCTION – SECURITY (ICAO an 17)

It is important to realize that security and safety are not the same issues. Security has to do with everything that might help to prevent crime on the Airside. The most relevant issue for you in this respect is the Airport Identity Card. Only persons issued with this card are authorized to be on the Ramp.

What are the rules and regulations with regard to this card?

LEARNING OBJECTIVES

After completing this lesson you will:

- Explain the differences between Security and Safety.
- Explain the importance of airside security.
- Explain why the Airport Identity Card is important.
- List the Rules and Regulations regarding the Airport Identity Card.

WHAT DO YOU THINK?

- In the past, security and safety were sometimes used to describe the same sort of rules and procedures. But that was the past! Times have changed dramatically!
- Nowadays security is getting more and more important, especially at Airports and Airsides.

QUESTION 21

What do you think is the correct definition of security?

a) Measurements taken to prevent accidents, injuries and damage to equipment in the workplace
b) The controlling of access, and the protection of property and aircraft from acts of vandalism, aggression and terrorism.

2.1.1 WHAT IS SECURITY?

In this course, security means to control the access to property and aircraft and protect them from acts of vandalism, aggression and terrorism. **(التخريب، العدوان، الارهاب)**

Safety refers to the prevention of accidents and injury, or damage to equipment in the workplace.

This lesson is about security.

Be aware: If you see 'strangers' at the Ramp, challenge them to identify themselves, and advise your supervisor or team leader. Security at the Ramp is also your responsibility!

2.1.2 AIRPORT IDENTITY CARD

The most well-known and visible security measure is the Airport Identity Card. The purpose of the airport ID is for the Airport Authorities to be able to enforce the Airport regulations and identify unauthorized persons on the airside.

By law, everyone who works in the Airside environment is required to carry an Airport Identity Card.

QUESTION 22

What do you know about the Airport Identity Card?

Once you have received the Card, it is officially yours.

- True
- False

QUESTION 23

If you violate the Airport Regulations, your card can be confiscated?

- True
- False

QUESTION 24

Giving wrong details and false information is a criminal offence ?

- True
- False

QUESTION 25

If you have to work on an aircraft, you have to hide away your card safely.

- True
- False

QUESTION 26

You are the only person allowed to use your ID card?

- True
- False

QUESTION 27

You are allowed to use your Airport ID card outside your airport job ?

- True
- False

2.1.3 AIRPORT ID RULES

You have probably answered all the statements correctly.

But never forget that the ID is issued by your airport and remains the property of the airport. It can be confiscated in case of misuse or violation of airport regulations. Giving false information in order to obtain an ID card is a criminal offence.

The following rules apply to the airport ID card:

- It must be displayed at all times when working on the airside.
- The ID card may only be used by you.
- The ID card may only be used when working. Using your ID for other purposes than doing your job can result in serious penalties.

2.1.4 TAKING PHOTOS

In most airports it is forbidden to take photos or videos without permission from the airport's Security Department.

QUESTION 28

When I forget to bring my ID, I should inform my supervisor and go back home to get it ?

- True
- False

QUESTION 29

When I stop working for the company, I can keep my ID card as a souvenir?

- True
- False

QUESTION 30

When I don't follow the airport regulations, I risk losing my ID card?

- True
- False

SUMMARY

Security means to control access to the airside and to protect property and aircraft from acts of vandalism, aggression, and terrorism.

Access to the airside is controlled by means of the Airport Identity Card. When working on the airside your airport ID should always be visible. You may not use you ID for any other purpose than doing your job. The ID remains the property of the airport.

To be aware of the airside security is everybody's responsibility, also yours! Inform your supervisor or team leader if you notice something you don't trust!

Section 2 Aircraft Danger Zones

INTRODUCTION - AIRCRAFT DANGER ZONES (IATA AHM 630)

In this lesson, some basic technical principles about aircraft will be introduced and explained. You will find out why you have to be very careful around aircraft engines and what the critical parts of an aircraft are.

LEARNING OBJECTIVES

After completing this lesson you will:

- Describe the Danger Zones around jet engines.
- List the forces acting on an aircraft in flight
- Define thrust
- Describe the two Danger zones around jet engines and explain why they exist
- Describe the dangers around propeller aircraft.
- Explain what the Anti Collision Beacon (ACB) is.
- Describe what you should and should not do around aircraft and their engines to maintain a safe working environment.

✍ QUESTION 31

What is the most dangerous period during the aircraft ground handling?

a) When the engines of an aircraft are about to start.
b) When loading equipment is being operated.

2.2.1 DANGER!

The most dangerous period in aircraft ground handling is when an aircraft has the engines running or is about to start the engines.

Aircraft have huge engines that power them. The two major types of aircraft engines are the jet engine and the propeller engine.

Both jet engines and propeller engines can easily kill someone if a person comes too close.

2.2.2 FOUR FORCES

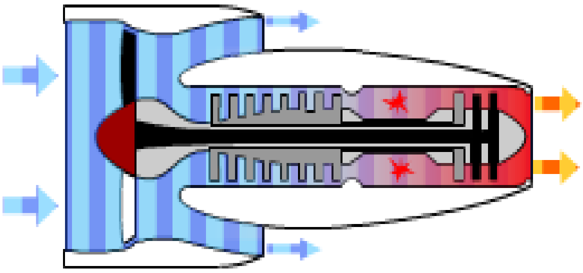
There are 4 forces acting on an aircraft in flight:

- Thrust
- Weight (Gravity)
- Lift
- Drag

Thrust is the force that moves an aircraft through the air. When the thrust is bigger than the drag the aircraft accelerates.

When this force is less than the drag, the aircraft will slow down.

2.2.3 THRUST



Thrust is generated as a result of accelerating a mass of gas. To produce the power that will create 'thrust' with a jet engine, air is drawn into the front intake of the engine. This creates a huge suction at the engine intake. Always be extremely careful around jet engines.

2.2.4 DANGER ZONES

There are 2 very important Danger Zones around jet engines.

Jet Ingestion: الابتلاع

At the front of the engines, air is drawn into the engine intake with a huge suction effect. This can be (and has been) strong enough to suck a human body into the engine intake.

Jet Blast: الانفجار

At the back of the engines, air is blasted out of the engine exhaust at high temperature and velocity. As thrust increases so does the temperature and velocity of the air.

QUESTION 32

At the sides of the jet engines the (lethal) suction effect is zero.

- True
- False

QUESTION 33

Make sure that loose equipment and/or clothing remains well clear of the Danger Zones

- True
- False

QUESTION 34

Always stay out of the Danger Zones when the Anti Collision Beacon is flashing.

- True
- False

DO'S AND DO NOT'S

Always be aware of where the Danger Zones are!

Never approach an aircraft until the aircraft engines are shut down, the Anti Collision Beacons (ACBs) are off, and the chocks are placed.

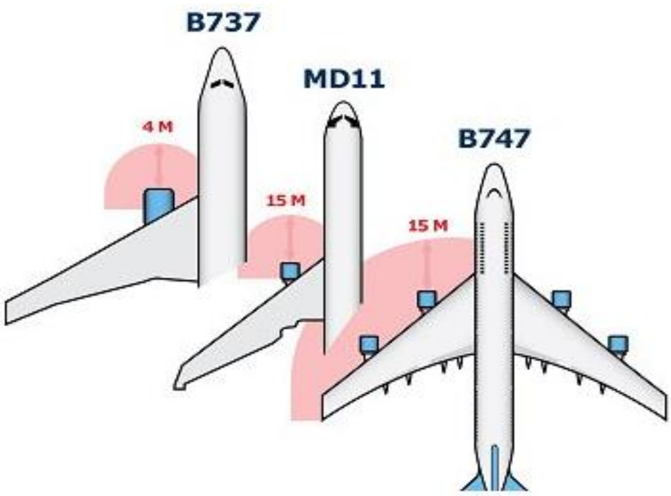
The suction effect at the jet ingestion (engine intake) is also significant at the side of the engine intake. If you come too close it can be lethal.

Always be careful with loose equipment and/or clothing. These items should always remain well clear of the Danger Zones when engines are running or about to start.

When the engines are running or about to start under no circumstances are you allowed to come close to the Danger Zones!

2.2.5 JET ENGINE INGESTION

- Jet aircraft can have different jet ingestion or engine intake Danger Zones. Be aware that these zones can reach as far as 15 m (50ft).
- The suction is also effective at the side of the engine intake. Never come too close
- Take a closer look at these different aircraft types and make sure that you know where the Danger Zones are.



2.2.6 JET ENGINE BLAST

- Air is blasted out of the engine exhaust at both high temperature and speed.
- The following table indicates the average heat and speed of the blast, and the clearance length when a jet engine is running at taxi power (idle) and at take- off (break-away):

heat	50 ^{0}C	120 F
speed	120 km/h	75 mph 65 knots
length (idle)	30 m	100 ft
length (take-off)	250 m	800 ft

As thrust increases so does temperature and velocity.

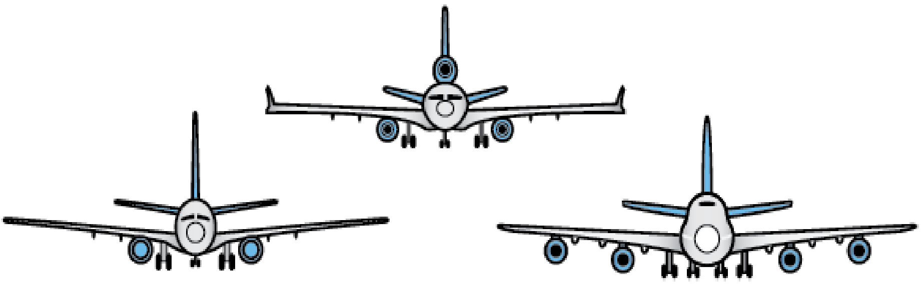
A “rule of thumb” would be:

- Idle Power: Clearance of at least one length of aircraft.
- Break-away Power: Clearance of at least two lengths of aircraft.

Don't get blown away!

2.2.7 ENGINE NUMBERS

- The engines are numbered in sequence from port to starboard, starting with the outermost engine on the port side, numbered 1.



2.2.8 PROPELLER AIRCRAFT

- Now you are aware of the dangers of jet engines. But what do you know about propeller aircraft?
- Propeller aircraft are particularly dangerous because as a propeller is turning at high speed it can become almost invisible to the eye.
- That is why everybody has to remain clear of the Danger Areas of the propeller(s) when the Anti Collision Beacon (ACB) is flashing.
- Nobody is allowed to walk through the propeller turning area during the turnaround as propellers can turn in the wind at any time and cause injury.

QUESTION 35

You should never stop a propeller by hand even when it is not turning very fast ?

- True
- False

QUESTION 36

Only when the propellers are stopped can you walk under the propeller blades ?

- True
- False

QUESTION 37

When propellers shut down they make a high pitched squealing noise ?

- True
- False

QUESTION 38

Turning propellers are not easily seen.

- True
- False

2.2.9 BE AWARE!

The following list indicates some important Do's and Do Not's with regard to propeller aircraft.

- Never attempt to stop the propeller by hand.
- Wait for propellers to come to a complete stop.
- Be aware that propellers are silent, especially after engine shut down.
- Turning propellers are not easily visible.
- Never walk in between or under the blades of propeller engines, even if the engines are stopped.
- When a propeller engine is running, the Danger Zone is 4 m (13 ft) around the propeller.
- Even when stopped, always go around the propeller path.

2.2.10 ANTI COLLISION BEACON

- The Anti Collision Beacon (ACB) has been mentioned before, but what is it?
- The condition of a running engine or an engine that is about to be started is signified by a flashing red warning light called an ACB, or Anti Collision Beacon.
- ACBs are located both on top of and below the fuselage of the aircraft.
- Whenever the ACB is on you must remain well clear of the Danger Zones.

SUMMARY

In this lesson you have learnt about the Danger Zones of jet engines and propeller engines.

Always remember:

Jet Engines:

- There are 2 Danger Zones: One at the front of the engine and one at the back.
- The Danger Zones are larger when the engines are in the break-away or take-off mode.

Propeller Engines:

- Even when the propeller is stopped, always go around the propeller path.

Anti Collision Beacons:

- Never enter the Danger Zones when the ACBs are flashing.
- Neglecting these rules can be life threatening. So always take good care!

Section 3 Health and Safety

INTRODUCTION - HEALTH AND SAFETY

- Awareness of the hazards, cautious behavior, and safety measures can reduce the risks involved with working on the Ramp.
- Personal protection Equipment (PPE) and High Visibility Clothing (HVC) are some of these measures. With High Visibility Clothing you stand out as much as possible from your surroundings.

LEARNING OBJECTIVES

After completing this lesson you will:

- List the Personal Protection Equipment (PPE) you must wear to protect yourself when working on the ramp.
- Describe how to prevent injury when:
  - lifting heavy objects
  - using the correct equipment
  - loading aircraft
- Define FOD and explain how it can lead to injury of personnel or damage to an aircraft.

2.3.1 HEALTH AND SAFETY

- Before setting foot on the Ramp you must be wearing protective footwear. If you wonder why, just try!
- Another hazard of working on the Ramp is not being seen by vehicle drivers or pilots. This has resulted in people getting seriously injured or killed. That is why you have to wear High Visibility Clothing.
- Continuous exposure to aircraft engine noise will seriously damage your hearing.
- In the following pages you will find out how to protect yourself and how to work without damaging your health.

2.3.2 PERSONAL PROTECTION EQUIPMENT

- To do your work safely, you are required to wear Personal Protection Equipment (PPE).
- This consists of:
  - High Visibility Clothing (HVC)
  - Protective footwear
  - Ear protectors
  - Work gloves
- You must be properly equipped before going on the Ramp.
- IT IS THE LAW!

QUESTION 39

Buy new PPE when the old equipment is damaged.

- True
- False

QUESTION 40

Check and report damages.

- True
- False

QUESTION 41

It is your responsibility to clean your PPE before and after use.

- True
- False

QUESTION 42

Store the PPE when you are not using it.

- True
- False

QUESTION 43

Clean the PPE before and after using it.

- True
- False

2.3.3 FACE PROTECTION

If there is a possibility of fluid 'splash back' when handling dangerous fluids you have to wear face protection. This could be in the case of de-icing or toilet servicing.

2.3.4 LIFTING

- A large percentage of personal injuries are caused by lifting objects. During the lifting operation the body can easily be forced off balance. There are several principles that should be applied to prevent muscle strain.
- Which of the following statements are true?

2.3.5 USE YOUR BODY

REMEMBER:

- Fully utilize the strong leg muscles for lifting rather than the comparatively weak back muscles.
- Use your body weight to initiate the movement in the desired direction.

- Maintain the natural shape of the spine.
- If the object to be lifted is too large or too heavy then seek assistance.

Never overstrain yourself. It may lead to serious injury.

2.3.6 USE THE EQUIPMENT

For performing activities above your head you can use stairs or steps. Sometimes hydraulic pumps can make heavy work easier as well. There is a lot of this type of equipment to make your work lighter, easier, and most of all, safer! Use it!

2.3.7 LOADING

If your work involves a lot of pallets, containers, and loose bulk cargo loading into the aircraft, keep the following rules in mind:

- Keep hands and feet away from stops/locks/guides.
- Never throw or drop loads but set them down easily.
- Slide rather than lift objects into place.
- Beware of trapping other personnel.
- Avoid handling load by the metal strapping.
- Beware of cargo tumbling out of carts when lowering cart gates.
- Always wear gloves

2.3.8 FOD

- FOD stands for Foreign Object Debris. FOD is any object that does not belong to an aircraft and can lead to injury of personnel and/or damage to the aircraft.
- It can be anything: a bolt, a concrete chip, a piece of paper, a hat, a tire tread, or even a passenger.
- FOD is found at terminal gates, cargo aprons, taxiways and runways. It causes damage through direct contact with airplanes, such as by cutting tires or being ingested into engines, or as a result of being thrown by jet blast and damaging airplanes or injuring people.

- Remember: It is everybody’s responsibility to avoid FOD on the Ramp and remove it when it is seen.

SUMMARY

You can reduce the risk of personal accidents and injuries considerably by following these rules:

- Use your Personal Protection Equipment (PPE).
- Use your body weight and leg muscles when lifting.
- Don’t bend your back, and seek help when an object is too heavy to lift!
- When moving around, operate vehicles and equipment in the intended way.
- Don’t throw or drop a load, and watch out for hands and feet.
- Use special equipment such as stairs, steps, or pumps to make the work easier.
- Never overstrain your body.
- Always pick up FOD when seen on the Ramp.

Section 4 Severe Weather Conditions

INTRODUCTION - SEVERE WEATHER CONDITIONS

- During severe weather conditions the airside can become a very dangerous working environment.
- That is why a special severe weather condition operations plan is put into action during these conditions.

LEARNING OBJECTIVES

After completing this lesson you will:

- List the specific weather conditions that require special actions.
- Describe each of the notification phases for lightning.
- Describe how a severe weather conditions alert is communicated.
- Describe how to proceed when
  - the visibility is less than 800 m
  - ground icing is present
  - there are high winds
  - moving baggage and cargo during severe weather conditions
  - there is lighting activity

2.4.1 SEVERE WEATHER CONDITIONS

- A Severe Weather Operations Plan will be activated during extreme weather conditions. There are 4 specific weather conditions that require special actions:
- High/sustained winds - Winds whether steady or gusting in excess of 75kph (40 knots).
- Lightning - The abrupt electric discharge from cloud to cloud as well as from cloud to earth
- Low visibility - As a result of heavy rain, snow, sandstorms, fog or conditions when visibility is typically less than 800m (0.5 mile).
- Ground icing conditions - This happens when there is snow and ice on surfaces and movement areas as well as when surface temperatures/wind chills can cause freezing.

2.4.2 EARLY WARNING

- When weather conditions become severe, a number of preventive measures need to be taken.
- In case of high winds these measures will require time and efforts so warnings at an early stage are particularly important.
- When an electrical storm with heavy lightning is detected within less than 5km (3 miles) of the operations, all the activities of the operating personnel should be stopped immediately.

2.4.3 LIGHTNING NOTIFICATIONS

- There are 3 notification phases for lightning:
  - Alert: Lightning activity is detected at a distance in excess of 8km (5 miles) from the operations.
  - Stop/Suspend Activities: Lightning activity is detected within 5km (3 miles) of the operations.
  - All Clear: Lightning activity has moved beyond 5km (3 miles) and is heading away from your operations.
- Note: These distances may vary depending upon local climatic parameters.

2.4.4 LIGHTNING ALERT

When lightning activity is detected at a distance of 8 km (5 miles) from the operations, a lightning alert will be given out.

The following should be done during the lightning Alert phase:

- Prepare for the STOP phase.
- Suspend non-essential activities in open areas.
- Reduce fuelling pressures to prevent the accumulation of static charges.
- Avoid using highly conductive equipment.

2.4.5 A 'SHUT DOWN' CONDITION

When lightning or an electrical storm comes closer (within 5 km or 3 miles) to the operations, all the activities of the operating personnel should be stopped immediately.

- Stop fueling.
- Interrupt aircraft communication by head set.
- Stop all ramp activity and clear area.
- Seek shelter inside buildings or inside metal bodied vehicles.
- Be aware that no one should seek shelter under any part of the aircraft, loading bridge, near light poles, fences, under trees

2.4.6 Passenger and boarding procedure:

- If passengers have not started boarding, they must be kept in the gate lounges.
- If boarding has started, stop the process and leave passengers already boarded in the aircraft.
- If an aircraft has just arrived it should be held off the gate until the lightning alert is removed.

2.4.7 GET THE WORD AROUND

- The following are the various methods used to communicate a severe weather conditions alert:

Radio communication.

- Visual signal:

  In some cases, various light colors will indicate the severity of the alert:

  - Red light = Take Shelter
  - Yellow light = Warning
  - Green light = All Clear
- When a single light system is used, a blue flashing light will indicate the alert to avoid confusion with other lights.
- Audio signal:

  Horns or sirens can be used. To be effective, the noise emitted by the horn or siren must be louder than the noise of the GSE and the aircraft engines.

QUESTION 44

During a thunderstorm right above the airside area:

Finish your activity as quickly as possible and make sure to use the right hand signaling.

- True
- False

QUESTION 45

All ramp activity must be stopped except the fuelling operation.

- True
- False

QUESTION 46

Finish your activity and seek shelter in a safe place.

- True
- False

2.4.8 LOW VISIBILITY

What should be done when the visibility is (or soon will be) less than 800m (0.5 mile)?

- Keep the minimum required equipment in the airside area. All other equipment should be removed.
- Reduce the equipment operating speed considerably.
- Turn on the lights on the motorized equipment.
- Take extra care at all intersections and vehicle/apron taxi-lane crossings.
- Obtain clearance from ATC (Air Traffic Control) prior to crossing taxiways (where permitted).
- Take additional care to ensure that vehicle windshields are clean.

2.4.9 GROUND ICING

- Both ground and work surfaces on equipment will become particularly hazardous during periods of ground icing conditions.
- When ground icing conditions are predicted, the equipment must be appropriately prepared to ensure its functionality and operational safety.
- Wherever possible, snow and ice formations on equipment and work surfaces should be removed prior to the start of operations.
- Personnel should allow extra time for activities, drive more slowly, and allow a greater distance to stop equipment.

2.4.10 SECURE AIRCRAFT DURING HIGH WINDS

During a high wind period the aircraft should be appropriately secured through a number of preventive measures.

The following is an overview of the preparations required.

- Additional chocks must be installed to ensure the aircraft does not move.
- All the doors, cockpit windows and panels must be closed.
- All nose gear torsion links must be secured.
- The tow bar must be hooked up, the tugs attached, and by-pass pins installed.
- The aircraft should preferably face the wind.

2.4.11 LOADING BRIDGES DURING HIGH WINDS

Just as for aircraft, loading bridges must be taken care of during high wind conditions:

- Retract ground power cords.
- Close all the aircraft doors, retract and lower loading bridges, and secure their wheels.
- Position loading bridges so that they face the wind, bring them closer to the terminal or tie them down.
- Remove any loose equipment, e.g. ladders, FOD, or containers.

2.4.12 GSE DURING HIGH WINDS

Remove non-essential ground support equipment from the airside area.

- Position essential equipment away from the aircraft and outside the path of possible aircraft movement.
- All equipment that is left outside must be secured with the brakes set.
- Disconnect strings of carts or dollies so each conveyance is held by its own brake or attach a vehicle to hold them in place.
- Ensure all containers are locked on dollies or transporters with doors or curtains secured.
- Remove all empty, loose containers from areas around aircraft. If possible tie them together or to a firm structure or store them indoors.
- Lower all high-reach equipment, e.g. loaders, steps, catering trucks etc. Remove any loose equipment, e.g. chocks, cones, ladders etc.

2.4.13 BAGGAGE AND CARGO

When the weather conditions are so severe that the aircraft must be secured and the GSE removed from the aircraft, the loading of the baggage must also be stopped.

The baggage room and cargo personnel also must be instructed not to continue loading or unloading the aircraft, If possible baggage rooms should be used to stow the luggage.

QUESTION 47

Place additional chock.

a) High Wing
b) Low Visibility
c) Lightning
d) Ground Icing

QUESTION 48

Only the minimum required equipment is permitted on the airside.

a) High Wing
b) Low Visibility
c) Lightning الصاعقة
d) Ground Icing

QUESTION 49

Stop the boarding process.

a) High Wing
b) Low Visibility
c) Lightning
d) Ground Icing

QUESTION 50

Take greater distance to stop equipment.

a) High Wing
b) Low Visibility
c) Lightning
d) Ground Icing

2.4.14 LIGHTNING SAFETY

During lightning activity personnel should not:

- Get out of enclosed vehicles.
- Use a headset connected to an aircraft.
- Use portable electronic devices, e.g. mobile phones, pagers, two-way radios in open areas or in front of windows.
- Stay in open areas or under aircraft.

- Seek shelter under a tall tree.
- Load or unload explosive or flammable material.

SUMMARY

- In this lesson, you have seen an overview of the severe weather conditions that can occur during operations on the airside.
- There should be a Severe Weather Operations Plan activated during High Winds, Low Visibility, Ground Icing and Lightning.
- Here is a summary of the appropriate procedures:
- During the 'shut down' condition, when there is Lightning within 5km (3 miles) of the operations, all the ramp activities should be stopped.
- During High Winds, aircraft, stairs, and bridges should be secured.
- Extra care should be taken with driving and placing GSE during periods of Low Visibility.
- During Ground Icing, snow and ice should be removed prior to the start of the operations.

Section 5 Traffic Rules at the Ramp

INTRODUCTION - TRAFFIC RULES AT THE RAMP

- The Ramp can be extremely busy with vehicles servicing the parked aircraft.
- This lesson is about the traffic rules at the Ramp.
- Even though you may not be driving a vehicle yourself, it still is important to be aware of the rules.

LEARNING OBJECTIVES

After completing this lesson you will:

- Describe how you obtain an Airside Driving Permit.
- List some traffic rules on the ramp.
- Explain where vehicles are allowed on the ramp
  - Define the Circle of Safety and the ERA
  - List the traffic rules for approaching an aircraft

2.5.1 AIRSIDE DRIVING PERMIT

In this lesson you will be introduced to the basic traffic rules. These rules apply for everybody working at the Ramp.

If you have to drive a vehicle on the airside you will need a special Airside Driving Permit. The rules, regulations, and terms for this Airside Driving License are set by the issuing authority, which is normally the local Airport Authority.

To obtain an Airside Driving Permit requires a separate airside vehicle driving training combined with a practical driving test.

2.5.2 DO YOU REMEMBER?

- The Apron is the area used for loading, unloading and parking aircraft. These 'parking spots' are usually called: Gates, Bays, or Stands.
- The roads that are used by vehicles for servicing Aircraft are called service roads. These roads are not accessible to aircraft. Aircraft can cross service roads at crossings controlled by traffic lights or stop bars.
- The combined area of Apron and service roads is called the Ramp.
- This lesson is about the traffic rules on the Ramp.

QUESTION 51

Pedestrians (e.g. passengers) are only allowed to walk on the Runways if they are escorted by authorized staff.

- True
- False

QUESTION 52

Servicing vehicles are allowed to drive between the Stands, only if there are no aircraft parked.

- True
- False

Note:

- Both of these statements are false. These actions can be seen as important Do Not's.
- Read about the Do's and the Do Not's with regard to Traffic and Driving on the Ramp in the following screens.

2.5.3 DO'S AND DO NOT'S

QUESTION 53

Can you judge these rules correctly?

a) Be aware of the turning radius of a dolly train.
- Do
- Do Not

b) Carry more passengers than seats in a vehicle.
- Do
- Do Not

c) Drive across empty stands.
- Do
- Do Not

d) Give way to aircraft (even under tow).
- Do
- Do Not

e) Obey signals and speed limits.
- Do
- Do Not

f) Pick up FOD seen on the Ramp.
- Do
- Do Not

g) Reverse a vehicle on the Apron, without a guide outside the vehicle.
- Do
- Do Not

h) Shut doors and shutters of vehicles.
- Do
- Do Not

i) Smoke inside an aircraft or vehicle.
- Do
- Do Not

j) Pay attention to pedestrians, passengers or staff.
- Do
- Do Not

k) Walk on the maneuvering area.
- Do
- Do Not

2.5.4 DO'S

- Obey all signs and speed limits.
- Always give way to aircraft (including those under tow).
- Always pick up FOD seen on the Apron.
- Shut vehicle doors and shutters.
- Take good notice of pedestrians, passengers or staff.
- Be aware of the turning radius of a dolly train.

2.5.5 DO NOT'S

- No Smoking on the Airside, including inside aircraft and vehicles.
- A vehicle should NEVER reverse on the Apron without a guide who must be outside the vehicle.
- Do not obstruct the emergency exit route of a fuel bowser.
- Carrying more passengers than there are seats in a vehicle is not allowed. NO SEAT = NO RIDE
- Pedestrians are not allowed to enter (walk on) the Maneuvering Area at any point, including the designated crossings.
- Vehicles must not be driven across aircraft stands, even if the stand is empty

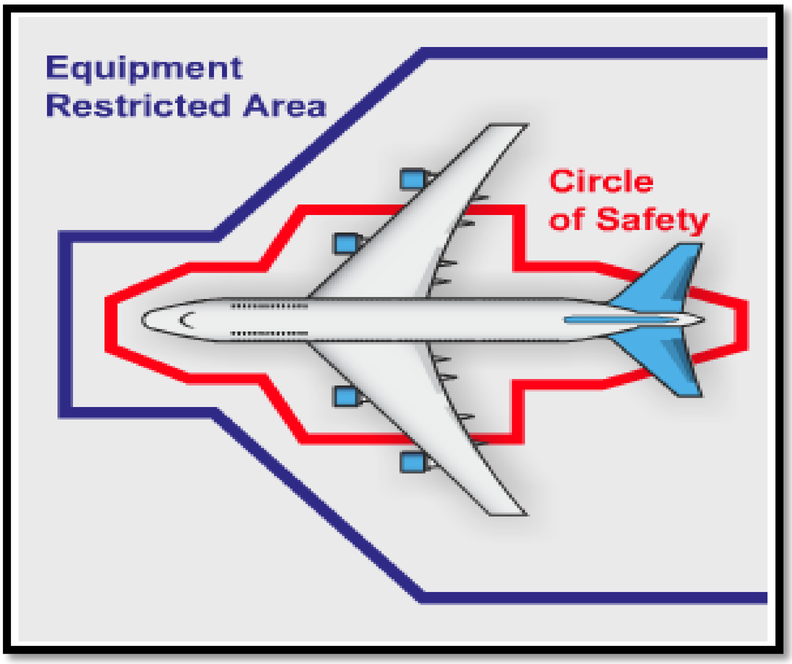
2.5.6 CIRCLE OF SAFETY AND ERA

When coming close to an aircraft, the “Circle of Safety” and the “Equipment Restricted Area (ERA)” should be adhered to.

These areas are marked by imaginary lines around the aircraft.

	Circle of safety	ERA
Nose	3m-10ft	7.5m-25ft
Fuselage	3m-10ft	---
Engines	3m-10ft	---
Tail	3m-10ft	7.5m-25ft
Wing tips	---	7.5m-25ft

When the engines are not running you must maintain at least the "Circle of Safety" around the aircraft.



<u>2.5.7 APPROACHING AN AIRCRAFT</u>

- When positioning a vehicle next to, or moving away from, an aircraft, speed must be at walking pace (6 k/h or 4 MPH).
- Only equipment that must be very close to the aircraft is allowed in the Circle of Safety.
- Only the equipment directly necessary to handle the aircraft can enter the ERA.
- All the other equipment is supposed to stay outside the ERA.

<u>2.5.8 PRIORITIES</u>

- Aircraft and pedestrians always have the right of way over vehicles and equipment.
- Vehicles and equipment should NEVER move across the path of the taxiing aircraft or embarking or disembarking crew.

<u>SUMMARY</u>

In this lesson you have been introduced to the most important traffic rules at the Ramp:

- If you are going to drive a vehicle on the Ramp you need a special Airport Driving Permit and you have to do a separate training and driving test.
- Be aware of the traffic at the Ramp. You now know what equipment is allowed in the Circle of Safety and in the Equipment Restricted Area (ERA).
- Aircraft and pedestrians always have right of way.

Section 6 Accident and Incident Reporting

INTRODUCTION - ACCIDENT AND INCIDENT REPORTING

Even though everybody tries very hard to create and maintain a safe environment for working on the Ramp, it is an unfortunate fact that accidents and incidents can, and do, occur.

LEARNING OBJECTIVES

After completing this lesson you will:

- Describe the reasons why you must report accidents and incidents at an airport.
- Explain the difference between an accident and an incident.
- List typical accidents and incidents that occur at the ramp.
- List report types and when they need to be completed following an accident or incident.

Which of the following statements are true?

QUESTION 54

You must always report accidents, and even minor incidents.

- True
- False

QUESTION 55

Being involved in an incident will lead to dismissal.

- True
- False

Often an incident is not reported because people are afraid of being dismissed. But you should be aware that failing to report an incident or accident is regarded far more seriously. It might therefore have more serious consequences.

غالبا ما يكون الحادث لم يبلغ عنها لأن الناس يخافون من الطرد. ولكن يجب عليك أن تدرك أن الفشل في الإبلاغ عن الحادث أو حدث يعتبر حتى الآن أكثر جدية. وبالتالي قد يكون لها عواقب أكثر خطورة.

2.6.1 LEGAL CONSEQUENCES

In most countries, and especially at most Airports, companies have a statutory obligation (a legal duty) to investigate and report accidents and incidents, especially those that involve serious injury or death, or involve an aircraft.

Failure to comply with statutory obligations could involve individual staff being held legally responsible for violation of these regulations. For instance, insurers could refuse to accept liability. In some countries, this may well result in the company or individual being held liable, which might result in your company or yourself having to pay for the damage.

Remember: Failing to report an accident or incident might cause a lot of problems!

في معظم البلدان، وخصوصا في معظم المطارات والشركات لديها التزام قانوني (واجب قانوني) للتحقيق وتقديم تقرير عن الحوادث والوقائع، وخاصة تلك التي تنطوي على إصابات خطيرة أو الوفاة، أو تنطوي على الطائرات.

يمكن عدم الامتثال للالتزامات القانونية إشراك الموظفين الفردية التي تتحمل المسؤولية القانونية عن انتهاك هذه القواعد. على سبيل المثال، يمكن أن شركات التأمين ترفض قبول المسؤولية. في بعض البلدان، وهذا قد يؤدي بشكل جيد في شركة أو فرد كائنا مسؤولا، الأمر الذي قد يؤدي في شركتك أو لنفسك الحاجة إلى دفع عن الضرر.

تذكر: الفشل في الإبلاغ عن وقوع حادث أو واقعة قد يسبب الكثير من المشاكل!

2.6.2 LESSONS LEARNT

- Reporting is not only critical for legal implications, but even more so for the “Lessons Learnt”.
- Reporting, even the smallest incidents, is important for the investigation. The main reason for reporting an incident is to identify how and why it happened, and to draw lessons from the incident.
- Next time effective action can be taken to prevent an accident and to reduce the risks.
- Remember: Reporting to prevent incidents in the future is also your responsibility!

- التقارير ليست حرجة فقط عن الآثار القانونية، ولكن حتى أكثر من ذلك من أجل "الدروس المستفادة".
- الإبلاغ، حتى أصغر الحوادث، من المهم للتحقيق. السبب الرئيسي للإبلاغ عن حادث هو تحديد كيف ولماذا حدث وما حدث، واستخلاص الدروس المستفادة من الحادث.
- يمكن أن تؤخذ في المرة القادمة إجراءات فعالة لمنع وقوع الحوادث والحد من المخاطر.
- تذكر: تقديم التقارير إلى منع وقوع حوادث في المستقبل هو أيضا مسؤوليتك!

2.6.3 INCIDENTS & ACCIDENTS

When you make a report a distinction has to be made between an incident and an accident.

- An incident is an event that disrupts the safety or order.
- An accident is an event resulting in injury or damage.

Many airports have special Emergency Phone Numbers to report incidents and accidents.

Make sure you know the emergency phone numbers at your airport.

- عند تقديم تقرير التمييز لابد من بذل بين حادث وقوع حادث.
- حادث هو الحدث الذي يعطل السلامة أو النظام.
- حادث هو حدث يؤدي إلى الإصابة أو الضرر.
- العديد من المطارات لديها أرقام الهاتف في حالات الطوارئ خاصة إلى الإبلاغ عن الحوادث والحوادث.
- تأكد من أنك تعرف أرقام هواتف الطوارئ في المطار الخاص بك

2.6.4 AT THE RAMP

- Typical accidents and incidents that occur at the Ramp include:
- Accidents involving personal injury to staff.
- Accidents involving Ground Support Equipment or other vehicles.
- Ground handling accidents/incidents.

- Aircraft damage.
- Aircraft emergency.

It is important to know the basic actions you have to take when you are involved in, or witness, one of these accidents or incidents.

2.6.5 FORMS

There are different types of reports that may be applicable when an accident or incident occurs. Some of these reports apply to international standards and regulations and some are specific to the company or local Airport Authority.

Report types include:

- Personal Injury Report. The form and procedures used may vary from station to station.
- Motor Vehicle Accident Investigation and Analysis Report.
- Aircraft Ground Handling/Incident Report. This is a report based on international standards (JAR OPS)

هناك أنواع مختلفة من التقارير التي قد تكون قابلة للتطبيق عند وقوع حادث أو واقعة.

بعض هذه التقارير إلى تطبيق المعايير واللوائح الدولية وبعضها محددة للشركة أو هيئة مطار المحلية.

وتشمل أنواع التقارير:

- تقرير الإصابات الشخصية. قد تختلف شكل والإجراءات المستخدمة من محطة إلى محطة.
- المركبات ذات المحركات التحقيق في الحوادث وتقرير التحليل.
- المناولة الأرضية للطائرات / تقرير حادث. هذا هو التقرير استنادا إلى المعايير الدولية (JAR OPS)

2.6.6 USING THE FORMS

- In some situations, all three forms must be completed. An example of this is when a motor vehicle causes injury to a staff member and damage to an aircraft.
- All staff members are required to report and assist their employer to the best of their ability in case of an accident investigation.
- It is the responsibility of supervisors to keep themselves informed about the applicable procedures related to completing the forms.

Are these statements True or False?

QUESTION 56

When an accident happens, three forms are always submitted.

- True
- False

QUESTION 57

When an accident happens, only the applicable forms are submitted.

- True
- False

QUESTION 58

When I have caused an accident, I should inform my supervisor about the details to the best of my ability.

– True

– False

SUMMARY

When an accident occurs it is important to follow the reporting procedures promptly and accurately for three main reasons:

- It is the law!
- If you fail to provide information, or give false information, you may have to pay for the damage yourself!
- A thorough investigation is necessary to learn from accidents and incidents and avoid damage or injury in the future.

An accident or incident is reported using one (or more) of three different forms:

- To report personal injury.
- To report the involvement of motor vehicles or GSE.
- To report the involvement of aircraft.

Part 3 Fire & First Aid

INTRODUCTION

- After studying this module you will know how to prevent fire when working on the Ramp and how to act when confronted with fire.
- You will also learn how to act when you have to deal with an injury resulting from an accident.

LESSONS

This module contains the following lessons:

- **Fire Prevention**: This lesson explains the best ways to prevent fires on the Ramp.
- **Fire Protection & Fire Action**: This lesson describes the theory behind the different types of fire and how they can be fought.
- **First Aid**: This lesson describes basic first aid procedures.

PRE-ASSESSMENT

The following slides are a pre-assessment to help you assess your own knowledge before taking the course. The questions will help prepare you for the material to be learned in this module. Your score is not recorded; it is purely for your own use for self-evaluation.

QUESTION 59

Smoking is prohibited on the ramp anyplace and anytime.

- True
- False

QUESTION 60

What fire class deals with paper, wood, and other combustible solids?

- A
- B
- C
- D

✍QUESTION 61

Who should your first call be to when you discover a fire?

a) Your supervisor
b) The Air traffic controller
c) The local fire department
d) The Airports Fire Service

✍QUESTION 62

If you are confronted with an accident with injured casualties, must you always call for an ambulance?

- Yes
- No

✍QUESTION 63

When arriving on the scene of an accident, what is the first thing you should do?

a) Assess if there are any dangers to you in approaching the accident scene.
b) Check to see if the victim is conscious.
c) Assess the situation, are there others more qualified to help?
d) Check to see if the victim is breathing.

Section 1 Fire Prevention

INTRODUCTION - FIRE PREVENTION

- Aircraft can carry a lot of fuel! This fuel is a potential fire risk.
- There are also other chemicals and fuel for GSE equipment involved during handling and turnaround of aircraft; another potential fire risk.
- But the biggest potential fire risk is people who are unaware!

LEARNING OBJECTIVES

After completing this lesson you will be able to:

- Explain how to prevent fire when working on the Apron.
- Describe precautions to take during aircraft refueling.
- List the rules of conduct regarding fire equipment.

WHAT DO YOU THINK?

✍QUESTION 64

Which of the following are the 3 main causes of fire at the Airside?

a) Accidents Yes No الحوادث
b) Arson Yes No الحريق
c) Carelessness Yes No لا مبالاة
d) Ignorance Yes No جهل
e) Lightning Yes No صاعقة
f) Short-circuiting Yes No الدائرة القصيرة

3.1.1 NO SMOKING

The most important and well known rule with regard to fire prevention at the Airside is:

- Never smoke on the apron area!
- Smoking or open fire is strictly prohibited anyplace and anytime!
- This also includes inside any vehicle or GSE!

3.1.2 GOOD HOUSEKEEPING

A clean and tidy Apron environment makes is easier to prevent a fire from starting, and to control it if it should occur.

Refuse or other combustible material must not be left to accumulate in buildings, on the Apron, or in vehicles.

Aircraft refuse must be collected and disposed of immediately. Be aware that your careful behavior might reduce the risk of fire.

3.1.3 NOT ONLY AT THE APRON

Don’t leave crates or any other packing material to accumulate in the warehouse or workshop. Dispose of it correctly in bins or containers.

Also, report any fault you notice with electric wiring immediately.

3.1.4 REFUELLING

There is an increased risk of fire during the refueling period. To prevent incidents during this period, you have to adhere to the following rules:

- Don’t obstruct the escape route for the fuel truck, and don’t park within the Refueling Safety Zone.
- Don’t position, start, (Dis) connect, or refuel any Ground Support Equipment (GSE).
- Don’t use camera flash or gas lanterns.
- Don’t operate electronic equipment such as radios, radar, or navigation equipment.
- Be Aware: The zone of 3 meters around or as specified by local regulations, from filling and venting points on the aircraft, fuelling vehicles and within the hydrant pits is called the Refueling Safety Zone. Especially within this zone you have to be extremely careful.

3.1.5 REFUELLING WITH STAFF ON BOARD

When there are people on board the aircraft during refueling, extra care must be taken:

- Keep all exits and aisles free, on board as well as on the ground.
- Passenger steps, passenger bridges, or integral stairs must be positioned properly for exit.
- On board staff must be informed when refueling is about to start.
- On board staff must be informed of any potential hazardous situations.

Remember to make sure that ramp staff is informed that boarding has started while fueling so that all emergency doors are kept free of obstacles (carts, ramp equipment, etc).

3.1.6 FIRE FIGHTING EQUIPMENT

Never obstruct access to equipment for fire detection, fire safety, and fire extinguishing. Make sure that all fire equipment is always visible. Don't obstruct vision in any way.

BE PREPARED

You may not be involved in risky activities yourself, but always be prepared to act in the correct way if a fire incident occurs.

Are you prepared?

Can you answer the following questions for yourself?

- Do I know where the Fire Fighting Equipment is?
- Do I know how to operate this equipment in case of an emergency?
- Do I know what the emergency number is and whom I should call?
- Do I know what the emergency procedures are?

If you cannot answer these questions, contact your supervisor.

Make sure you know the answers before entering the Ramp!

NOW YOU KNOW

Fire prevention is all about:

- Knowing the rules and procedures.
- Acting accordingly.
- Being very careful when working on the Ramp.

Now you should know whether the following actions are Correct or Incorrect:

QUESTION 65

Fuelling is never allowed as long as there are people on board.

- Correct
- Incorrect

QUESTION 66

I will leave it to others to call the Fire Service when necessary.

– Correct
– Incorrect

QUESTION 67

I cannot use the GPU during fuelling if it is within 3 m (10 ft) of the fuel truck.

– Correct
– Incorrect

QUESTION 68

I have to make sure that I always clean up refuse, even if it is not mine.

– Correct
– Incorrect

QUESTION 69

I have to know what to do in case of an emergency.

– Correct
– Incorrect

SUMMARY

Fire can be prevented if carelessness, ignorance, and accidents are reduced. To improve careful behavior, and to reduce ignorance, everybody working on the Ramp should work according to the following rules:

Don't smoke anywhere on the apron area.

- Leave your work place clean and tidy.
- Don't obstruct your view of, or access to, the fire equipment.
- During refueling don't operate any equipment within the 6 m refueling zone, and don't block the escape route for the fuel truck or for onboard staff.
- Be aware: If you need to perform tasks that involve fire risk, report this beforehand to the Airport Fire Service.

Now that you know how to prevent fire, proceed to the next lesson.

Section 2 Fire Protection & Fire Action

INTRODUCTION - FIRE PROTECTION AND FIRE ACTION

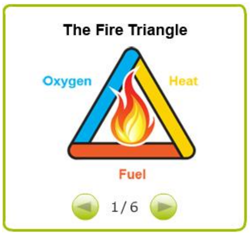
- If you discover a fire at the Airside you must inform the Airport Fire Service immediately.
- Although, extinguishing fires at the Airside is a job for specialized Fire Fighters, it is important for you to know and understand the basic principles of fire, how it can occur, and what it takes to extinguish a fire. It is also important that you know what you have to do when you discover a fire or a hydrant leakage.

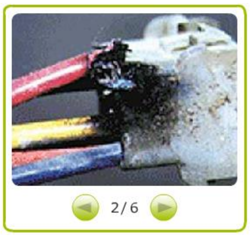
LEARNING OBJECTIVES

After completing this lesson you will:

- Describe the components of the Fire Triangle.
- List the different fire classes.
- Describe which extinguishers are used to combat different classes of fire.
- Explain what to do when a fire or a hydrant leakage is discovered.
- Describe the actions to take in case of an on board fire.
- Describe the actions to take in case of fuel spillage.

Remember: When a fire does not appear to be immediately controllable, the most important thing is to always call the Airport Fire Service!





WHAT DO YOU THINK

QUESTION 70

Indicate which of the following actions is the first thing you should do in case of fire?

a) When I discover a fire I should first try to extinguish it.
b) When I discover a fire I should first see if anyone must be safeguarded.
c) When I discover a fire I should first call the Airport Fire Service.

Correct.

The first thing is always to call the Airport Fire Service. They are specialized in extinguishing fire and safeguarding people.

For your general awareness some basic fire knowledge will be presented here.

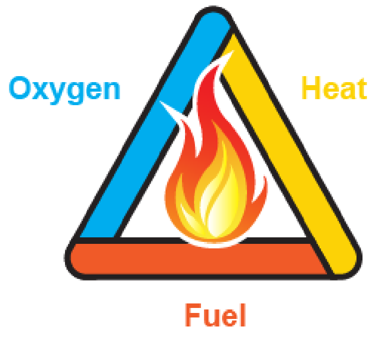
3.2.1 THE FIRE TRIANGLE

For any fire (combustion) to take place there must be three vital components present:

- Fuel: This can be solids, liquids or gases.
- Oxygen: This is present in the air.
- Heat: This can be provided by, for example, a cigarette, a lighter, the sun, or machinery parts getting hot.

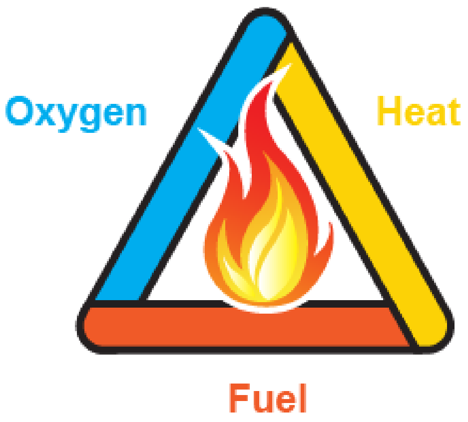
Fuel, oxygen, and heat are referred to as the "Fire Triangle".

The (perfect) combination of these 3 elements will result in the fourth element; a chemical reaction. And then there is a Fire!



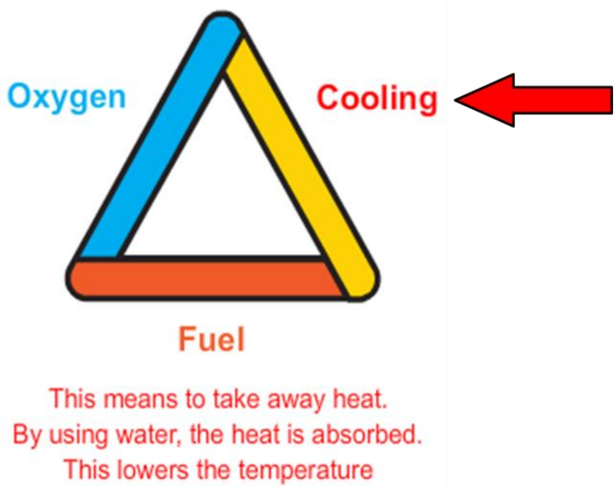
3.2.2 EXTINGUISHING

The important thing to remember is when you take any of these elements away, you will not have a fire, or the fire will be extinguished.

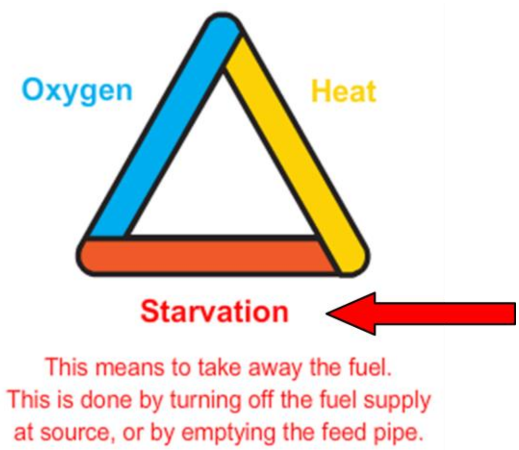


3.2.3 EXTINGUISHING

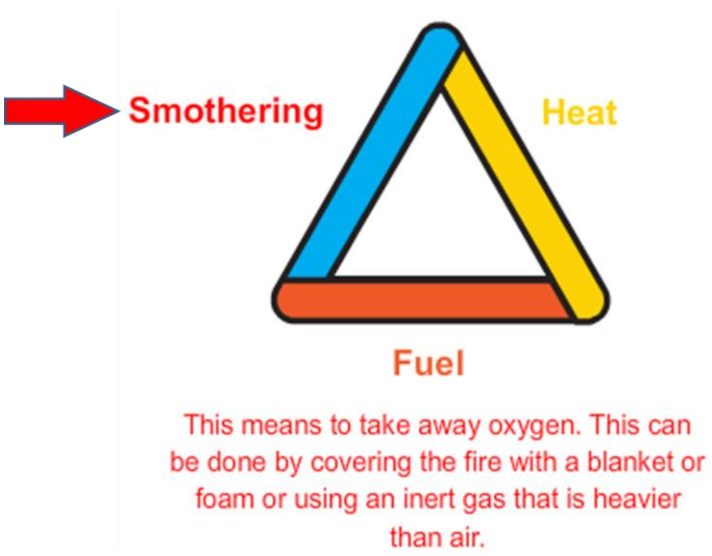
3.2.3.1 (Cooling)



3.2.3.2 Starvation

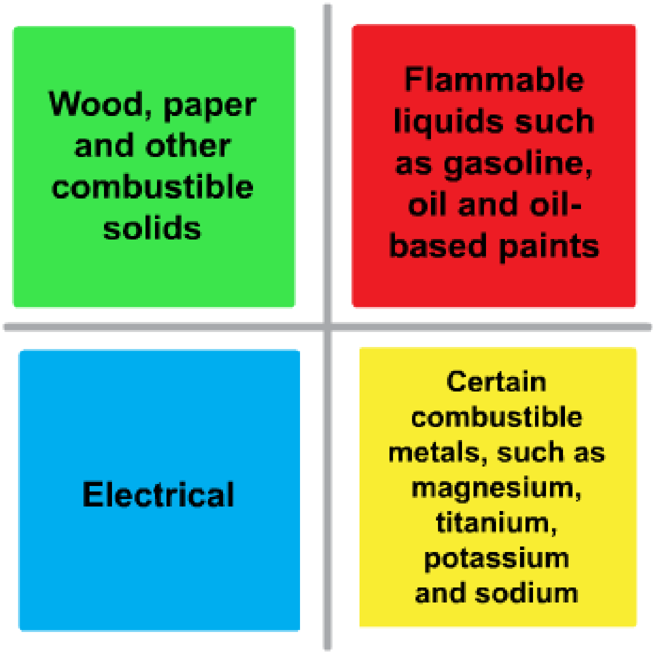


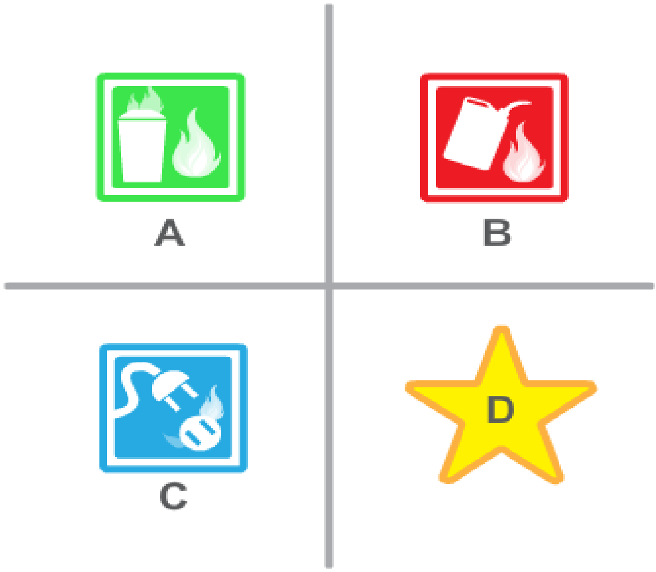
3.2.3.3 Smothering



3.2.4 CLASSIFICATION OF FIRES

- Because every fire is different, it is important to know how you should extinguish the fire. Therefore, different types of fires fall into different classes.
- Fire extinguishers are classified by letters according to the class of fire they can put out.
- Now you know that you can extinguish a fire in different ways.





3.2.5 ELECTRICAL FIRES

- An electrical fire is started by electricity. It can result in a class A, B or D fire.
- When an electrical fire occurs, the first action to take is to cut off the electricity, where possible!
- Remember: Never throw water on an electrical fire.
- When the electricity cannot be switched off, special extinguishers will be used.

NOW YOU KNOW!

QUESTION 71

When there is a fire, what should be done to extinguish it?

a) Use the type of extinguisher that suits the type of fire that is occurring

b) Wet the fire and the surrounding area thoroughly.

Correct

Fire can appear in many different forms and circumstances.

Depending on the type of fire the most appropriate extinguishing method should be used. And that could quite likely be something else than only water.

3.2.6 CALL FIRE SERVICE

When you discover a fire or a major hydrant leak, the first thing to do is call the Airport Fire Service.

Make sure you know the Emergency Phone Number of your Airport Fire Service.

When calling, give your exact location and the class of fire.

3.2.7 EMERGENCY ALERTING SERVICE

At some airports there are Emergency Alerting Services at the head of each stand. They include an emergency button to switch off the fuel and an emergency phone. Beside the phone you will find a description of the full location address.

Make sure to know the location of the Emergency Alerting Services points and read the instructions beforehand.

3.2.8 FIRE ACTION

You are on the right track now with your fire action.

You have called the Airport Fire Services.

What can you do now?

- Endeavour to safeguard the lives of passengers and colleagues, as well as the aircraft equipment involved.
- Only use the Fire Fighting Equipment if you are sure that you are using the correct extinguisher.
- Never tackle a fire if you cannot control it, or if it means putting yourself in danger!

3.2.9 ON BOARD FIRE

When there is a fire inside the aircraft you should first alert the Airport Fire Services and evacuate the cabin.

3.2.10 FUEL SPILLAGE

Action should also be taken in the case of fuel spillage or leakage. First things first!

What do you think has to be done first?

First read the following actions carefully, and then indicate the order in which actions should take place.

3.2.11 FUEL SPILLAGE

- Restrict all activities in the Area. 1 up to 6
- Evacuate all persons from the immediate area. 1 up to 6
- Switch off all equipment. 1 up to 6
- Do not resume normal operations until approved by the Emergency Service. 1 up to 6
- Call the Airport Fire Service. 1 up to 6
- Make sure all fuel supply is cut off (by informing the fueller or using the head of stand cut-off) 1 up to 6

SUMMARY

Be aware a small fire or fuel leakage at the Airside can have devastating results.

If there is a small fire, make sure that you can stop it by switching off the electricity or by putting the fire out with water or a fire extinguisher.

Never tackle a fire if you cannot control it, or if it means putting yourself in danger! Call the Airport Fire Service and give the exact location of the fire or leakage, as well as the type of fire if known.

- (If applicable) switch off the fuel at the Emergency Alerting Service point (Head of Stand fuel Cut-off).
- (If applicable) evacuate the cabin and crew and safeguard lives of persons involved without endangering yourself.
- In the case of a controllable fire, fight the fire with extinguishers.

Section 3 Fire Aid

INTRODUCTION - FIRST AID

First Aid is the care given to a casualty before professional help arrives.

First Aid can be as simple as putting a band-aid on a minor injury or as “difficult” as applying chest compression by trained First Aiders. The help you give can literally mean the difference between life and death

LEARNING OBJECTIVES

After completing this lesson you will:

- Define First Aid and list the aims of First Aid.
- List the actions to be taken when confronted with an accident with injured casualties.
- Describe what is meant by DR ABC.
- Describe the actions to be taken if a victim is unconscious.
- List the steps to take when a victim is bleeding heavily.
- Explain what shock is, how to recognize it in a victim, and what actions should be taken.
- List the actions to take with a burn victim.
- Describe common eye injuries and how to treat them.

WHAT DO YOU THINK

This lesson is about First Aid. Actions such as resuscitation and cardiac massage should only be carried out by trained First Aiders. However, there are enough other things you can do to help an injured person.

This module will introduce you to the basic principles of First Aid. You can learn enough First Aid in 20 minutes to save a life.

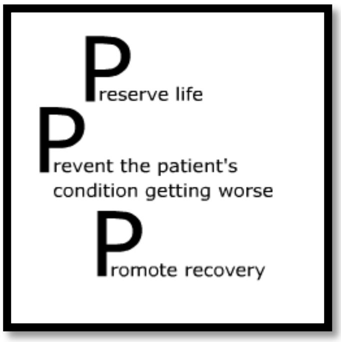
QUESTION 72

Which of the following statements is correct?

a) The best way to treat bleeding is to wash out the wound with a lot of water.
b) If you put a bleeding wound under a tap, you will wash away the clotting agents and make it bleed more.

3.3.1 DEFINITION

First Aid is immediate action taken by persons first on the scene of an accident, before professional help arrives.



The aims of First Aid are to:

- Preserve life.
- Prevent the patient's condition from getting worse.
- Promote recovery.

3.3.2 ACCIDENT WITH INJURED CASUALTIES

When you are confronted with an accident with injured casualties, you should take the following actions:

- Call for help immediately.
- Assess the scene.
- After checking the site is safe, assess the casualties and decide who needs help first.
- Determine whether the assistance of a trained First Aider is required, and request it if necessary.
- If required, make sure that the Ambulance Service is notified.

Remember: In case of a fall from heights, you must always notify the Ambulance Service.

3.3.3 INJURED PERSON ON THE FLOOR

- Imagine an injured person is lying on the ground. There is skilled help on the way.
- Complete the description below by selecting the correct actions to take in each drop down menu.

QUESTION 73

Always remember the general rule; if there is no evident threat to life, you should _______________ move the victim.

- Never
- Always

QUESTION 74

With an electrical accident _____ take further risks which may result in additional casualties.

- Do
- Do not

QUESTION 75

Switch off the electricity or, if that is not possible, ___________

- Drag
- Move

QUESTION 76

The casualty from the live system using a _____________

a) Dry wooden pole
b) By hand

3.3.4 ARE YOU A DR. ABC?

There is an easy reminder to help you remember what you have to do when giving First Aid.

This is called DR. ABC!

Find out what these letters stand for!

Once you know the meaning you can always act as DR. ABC.

3.3.4.1 DR,ABC

D = DANGER

Assess first whether the situation is safe for you. There might be moving traffic, chemical spills or other hazards.

R = RESPONSE

Check all sorts of things about the condition of the victim, by checking the responses. Ask; Are you all right?

If there is no answer start moving on with the ABC.

A = AIRWAY

Make sure the airway is clear for breathing. If in doubt, pull the jaw forward and the head backward.

B = BREATHING

If the breathing has stopped, start mouth-to-mouth resuscitation by pinching the nose and blowing in the mouth.

C = CIRCULATION

For this, you apply simple visual checks that the casualty's blood is circulating adequately, by watching for improved colour, for coughing or eye movement. If no pulse can be felt, cardiac massage must be applied. Trained First Aiders may only do this.

3.3.4.2 DR, ABC

- **D : Danger خطر**
- **R : Response استجابة**
- **A : Airway مجري الهواء**
- **B : Breathing التنفس**
- **C : Circulation تداول**

3.3.4.3 KNOW YOUR ABC

- D = Danger. Assess first whether the situation is safe for you. There might be moving traffic, chemical spills or other hazards.
- R = Response. Check all sorts of things about the condition of the victim, by checking the responses. Ask "Are you all right?" If there is no answer start moving on with the ABC!
- A = Airway. Make sure the airway is clear for breathing. If in doubt pull the jaw forward and the head backward.
- B = Breathing. If the breathing has stopped start mouth-to-mouth resuscitation by pinching the nose and blowing in the mouth.
- C = Circulation. For this, you apply simple visual checks that the casualty's blood is circulating adequately, by watching for improved colour, for coughing or eye movement. If no pulse can be felt cardiac massage must be applied.

Resuscitation and cardiac massage should only be carried out by trained First Aiders.

- **D = Danger.** Assess first whether the situation is safe for you. There might be moving traffic, chemical spills or other hazards.
  - تقييم أول ما إذا كان الوضع آمنا بالنسبة لك. قد يكون هناك تحريك حركة المرور، وانسكاب المواد الكيميائية أو غيرها من المخاطر.
- **R = Response.** Check all sorts of things about the condition of the victim, by checking the responses. Ask "Are you all right?" If there is no answer start moving on with the ABC!
  - تحقق من كل أنواع الأشياء عن حالة الضحية، عن طريق التحقق من الردود. نسأل: "هل أنت بخير؟" إذا كان هناك أي جواب على بدء التحرك مع ABC !
- **A = Airway.** Make sure the airway is clear for breathing. If in doubt pull the jaw forward and the head backward.
  - تأكد من أن مجرى الهواء واضح للتنفس. إذا كنت في شك سحب الفك إلى الأمام ورأسه إلى الوراء .
- **B = Breathing.** If the breathing has stopped start mouth-to-mouth resuscitation by pinching the nose and blowing in the mouth.
  - إذا توقف التنفس بداية من الفم للفم عن طريق معسر الأنف ونفخ في الفم.
- **C = Circulation.** For this, you apply simple visual checks that the casualty's blood is circulating adequately, by watching for improved colour, for coughing or eye movement. If no pulse can be felt cardiac massage must be applied.

- لهذا، يمكنك تطبيق الضوابط البصرية البسيطة التي الدم المصاب ينتشر على نحو كاف، من خلال مشاهدة لتحسين اللون، لاسعال أو الحركة العين. إذا يمكن أن يرى أي نبض يجب تطبيق تدليك القلب.

3.3.5 UNCONCIOUSNESS

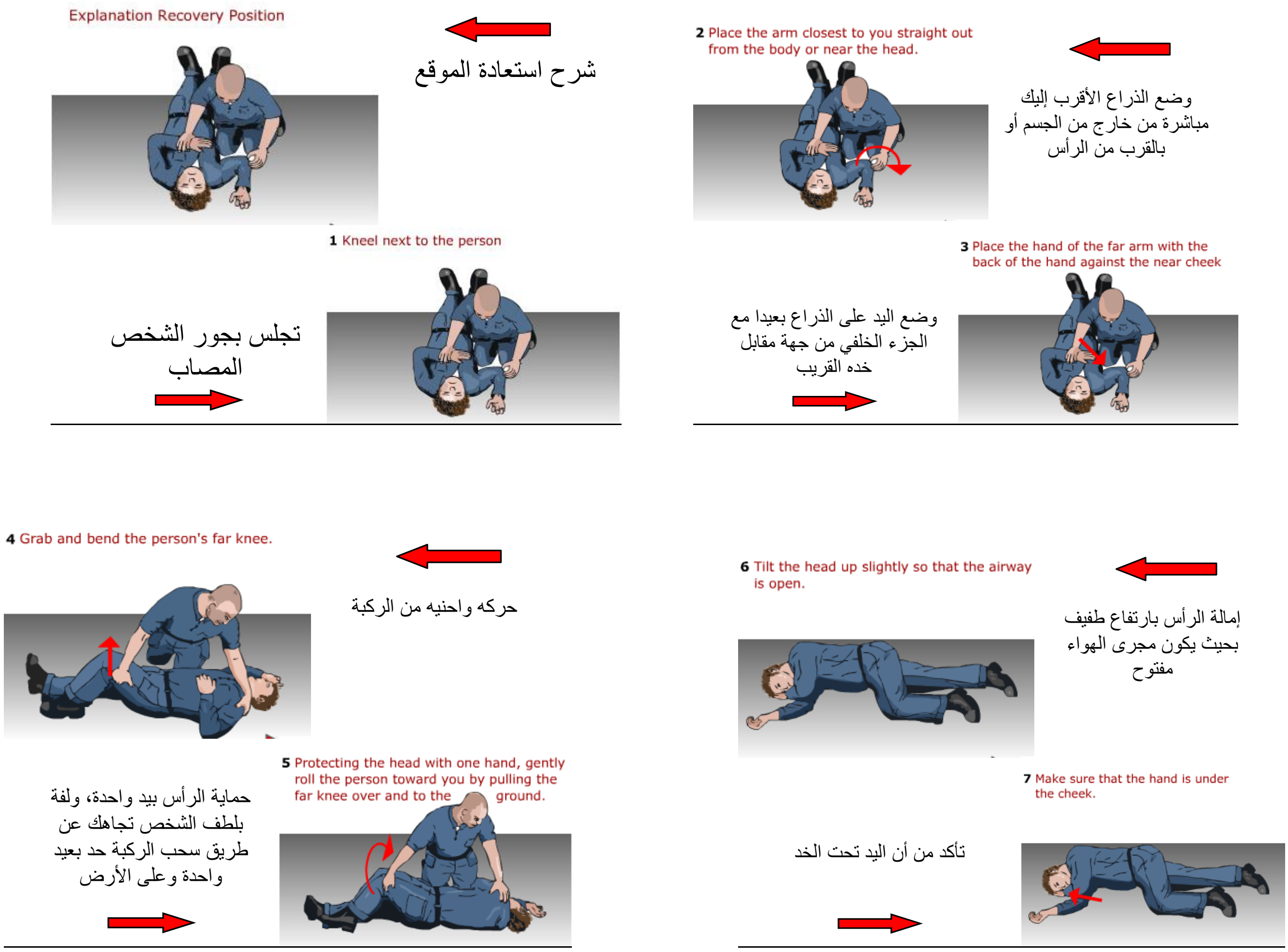
If the victim is unconscious they should be placed in the 'recovery position'. This position helps a semiconscious or unconscious person breathe and permits fluids to drain from the nose and throat so they are not breathed in.

Turn the victim on one side with the upper knee drawn up and the head turned to the same side.

- Check the airway (remove false teeth or any other obstruction).
- Do not try to give anything by mouth.
- Check for dangers such as fumes, fire or electrical shock.

إذا كانت الضحية فاقدا للوعي أنها ينبغي أن توضع في 'وضع الإفاقة. هذا الموقف يساعد الشخص شبه فاقد الوعي أو اللاوعي التنفس ويسمح لاستنزاف السوائل من الأنف والحلق بحيث لا يتم نفخ فيه أنها تحويل الضحية على جانب واحد مع الركبة العليا وضعت وتحول الرأس إلى الجانب نفسه.

تحقق مجرى الهواء (إزالة الأسنان كاذبة أو أي إعاقة أخرى) لا تحاول إعطاء أي شيء عن طريق الفم. التحقق من وجود أخطار مثل الأبخرة، حريق أو صدمة كهربائية.



WHAT DO YOU THINK?

If someone is bleeding heavily what should you do?

First read the following descriptions and choices carefully, and then select the appropriate action.

✍QUESTION 77

If someone is bleeding heavily what should you do?

a) Stop the bleeding.
b) Take away the cause for the bleeding.
c) Get them to the hospital as soon as possible.

✍QUESTION 78

How are you going to control the bleeding?

a) Wash the wound out with water.
b) Try to close the wound by using your hands.
c) Wrap a bandage around the wound.

✍QUESTION 79

What complications can a victim suffer after severe bleeding?

a) Shock
b) Panic attack
c) Breathing difficulties.

3.3.6 BLEEDING

The primary aim with a victim who is bleeding heavily is to minimize the bleeding.

- You should try to close the wound by using your hands.
- Push the sides of the wound together with your hands, keeping them as far away from the actual wound as possible.
- You have to apply pressure to stop the bleeding. Use a clean dressing if possible, but do not waste time looking for one.
- Bleeding limbs should be raised.

3.3.7 SHOCK

Shock can happen when the body diverts blood form the skin to the vital organs in an attempt to keep them functioning normally.

The casualties' skin may turn pale, grey, and sweaty. They may have a rapid pulse and develop nausea.

The best thing to do is to lay the victim in a position with his legs higher than his head. This will promote the flow of blood to the brain. Keep the victim warm.

Do not give the victim anything to eat or drink until approved by a doctor.

WHAT DO YOU THINK?

✍QUESTION 80

Someone has burnt his hands. What should you do?

a) Calm the person down and make him breath regularly.

b) Cool the burns down with water.

c) Give the burns as much fresh air as possible.

✍QUESTION 81

After the burn has cooled, what do you do?

a) Remove jewellery like watches and rings.

b) Place the victim in the recovery position.

c) Apply ointment or hand cream.

✍QUESTION 82

What would be a good dressing to cover the burn?

a) Clean towel.

b) Tissue.

c) Sterile bandage.

3.3.8 BURNS

If someone has been burnt:

- Hold the burnt skin under cold running water for up to 15 minutes.
- Remove anything which may cause discomfort (such as jewelry) if swelling occurs.
- Never use a cream or ointments!
- Apply a clean dressing, a sterile bandage, and seek urgent medical advice.

3.3.9 EYE INJURIES

The most common eye injuries are:

- Contamination with chemical splashes.
- Injuries from foreign bodies (small pieces).

In both cases you can take the following action:

- Wash the eye out with water for at least 10 minutes.
- Lay the victim down with their head supported.
- Get the victim to close their eyes and keep them still.
- Use a sterile pad as a dressing until the help arrives.
- Always seek medical advice immediately.

What should you do in the following circumstances?

First read the following descriptions and choices carefully, and then select the appropriate action.

NOW YOU KNOW!

- When someone is in shock I should:
  - Raise his legs and keep him warm.
  - Give him something to drink and keep him warm.
  - Get him to sit up and keep him warm.
- When someone is unconscious I should:
  - Get him to sit up and keep him warm.
  - Turn him carefully on his side and check breathing and pulse.
  - Apply cold running water to wake him up.
- The ‘recovery position' is to:
  - Keep the tongue and fluids from blocking the airways.
  - Make the victim feel more comfortable.
  - Make it easier to do mouth-to-mouth resuscitation.

SUMMARY: ALWAYS!

Always act in the following order:

- Call professional help and the Ambulance Service.
- Check for dangers such as fumes, fire or electrical shock.
- When the victim is in contact with an electric current, switch off the electricity.
- Find a trained First Aider.
- Make sure the victim's airway is clear for breathing.
- If breathing has stopped, start mouth-to-mouth resuscitation.
- In case of bleeding apply pressure to stop the bleeding.
- In case of a burn injury apply cold running water.

SUMMARY: NEVER!

Never:

- Move the victim unless there is direct danger.
- Give anything to drink or eat in case of shock or unconsciousness.
- Attempt a rescue when there is a high voltage.
- Apply cream or ointments on burns or wounds.
- Enter a fume filled area without protection if you don’t know what fume is released.
- Attempt cardiac massage when you are not trained to do so.

QUESTION 83

What is the most important and well known rule for fire prevention on the airside?

a) Refuel carefully

b) Don't refuel during lightning activity

c) Never smoke on the apron areas

d) Smoke only inside GSE

QUESTION 84

What type of extinguisher would you use to extinguish a gas fire?

- A
- B
- C
- D

QUESTION 85

When calling Airports Fire Service, what information do you need to give to the dispatcher?

a) Location and type of fire

b) Location and time of day

c) Suspicions of arson and type of fire

d) Suspicions of arson and location

QUESTION 86

If you encounter an accident where the victim has fallen from a height, you must always notify the Ambulance Service.

- True
- False

QUESTION 87

What does the 'B' mean in the acronym DR ABC?

a) Be aware
b) Bed rest
c) Be prepared
d) Breathing

Part 4 Safety Management

Section 1 Introduction of Safety Management Systems

FOREWORD

Safety Management Systems (SMS) represent a systematic, explicit and comprehensive process for managing risks to safety. Each system is based on the airport operator's in-depth knowledge of its organization, and integrates safety into policies, management and employee practices, as well as operating practices throughout the organization. As each organization integrates safety into daily operations, management and employees can continuously work to identify and overcome potential Safety hazards that could cause accidents.

Airport safety management systems are very specific to their particular industry segment and must allow all the airport stakeholders to interact in a joint effort to improve safety. An SMS has to be modular and commensurate with the airport size and operations. It has to be practical and efficient so as to ensure a high level of safety and not become counterproductive.

Airports play a crucial role in ensuring operational safety at their specific location. In most cases, the airport authority is responsible for the safety of all aeronautical operations taking place on their territory and in the surrounding airspace. For this reason, the airport SMS is a key component, which will greatly enhance the overall safety of civil aviation.

In addition to the requirements of *ICAO Annex 14 – Aerodromes, Volume 1, Chapter 1.5 (Safety Management*), ICAO has also published a very comprehensive *Safety Management Manual (Doc 9859)*, intended to provide States with guidance to provide the regulatory framework and the supporting guidance material for the implementation of an SMS by service providers.

The responsibility for the implementation of an SMS lies with all line managers and employees. Organizations may also have a specifically designated safety manager who monitors and assists in the implementation and audits compliance. Depending on the size and complexity of the organization, the safety manager function can either be a dedicated position, or include other responsibilities. Some organizations may also create a Safety Office that will be responsible for the implementation and development of the SMS.

Section 2 Sudan Civil Aviation Regulations

SUCAR Part 19 SAFETY MANAGEMENT

FOREWORD

SUCAR Part 19 – Safety *Management* has been promulgated pursuant to Article 33 of the Sudan Civil Aviation Act, 2010 and issued by the Board of Directors of Civil Aviation. The SUCAR has been published under my Authority on the advice of the Board of Directors of Civil Aviation as is required by the Sudan Civil Aviation Act.

The Standards contained in this document fully comply with the Standards and Recommended Practices (SARPs) contained in Annex 19 – *Safety Management* to the *Convention on International Civil Aviation,* signed in Chicago on 7 December 1944 (Chicago Convention) and related aircraft operation documents and guidance material issued by ICAO.

This First Edition of SUCAR Part 19 – *Safety Management* is designed to contain comprehensive regulatory requirements for the identification of hazards and the assessment and management of safety risks of all activities related to the operation of aircraft in the Republic of Sudan.

This SUCAR represents one of the principal regulatory requirements that would, in addition to the effective implementation of the other SUCARs, enable Sudan to effectively and efficiently implement its National Safety Programme and fully meet its obligations under the Convention on International Civil Aviation.

This SUCAR, in addition to its implementation under the Sudan Civil Aviation Act is also mandated by the National Safety Program, May 2015. As such, SUCAR Part 19 – *Safety Management* is fully supported by and supports all other safety-related SUCARs published by the Government of Sudan to enhance aviation safety that may also contain requirements for the management of safety in their specific areas of activity.

The effective and sustainable implementation of this SUCAR shall be supported by Guidance Materials, Operational Directives and Safety Notices that may be issued by the Director General of Civil Aviation as required, from time-to-time.

Section 3 Sudan Civil Aviation Authority

NATIONAL SAFETY PROGRAMME

FOREWORD

In response to the recommendations of the Directors General of Civil Aviation Conference on Global Strategy for Aviation Safety (Montreal, 20 to 22 March 2006), and the High-level Safety Conference (Montreal, 29 March to 1 April 2010 (HLSC/2010)), the International Civil Aviation Organization (ICAO) has developed and adopted provisions for the establishment of State Safety Programs (SSP) and also for the management of safety by service providers. Those provisions are contained in Annex 19 – *Safety Management* to the Convention on International Civil Aviation and are intended to assist States to manage aviation safety risks and support the continued evolution of a proactive strategy to improve safety performance. Obviously, the foundation of this proactive safety strategy is based on the establishment and implementation of a State safety program required by ICAO Annex 19 – *Safety Management* provisions.

Since 2011, the Republic of Sudan has made significant progress in the enhancement of aviation safety in the Sudan in general, and of its aviation service providers wherever they may be operating. This progress has been noted and recognized by ICAO and the international aviation community and has been quoted as an exemplary in various international and regional conferences and meetings. Sudan was able to achieve safety, enhancement through the establishment and management of an effective and sustainable national safety oversight system, recognized by the ICAO Universal Safety Oversight Program (USOAP) to be among the most improved in Africa with very high percentage of effective implementation of the critical elements of a safety oversight system.

Section 4 Risk Assessment and Control

Risk assessments allow airport operators to develop an objective assessment of the risk associated with a specific activity. Risk assessments should be conducted for every task to be carried out by staff and can also be carried out on a higher level of the operational business, for example concerning bird strikes or runway incursions.

Different countries will have separate requirements depending on their local legislation and environment. The 5 simple steps listed below are the basis of a risk assessment as outlined by the Health and Safety Executive (HSE):

1. Identify the hazards
2. Decide who may be harmed and how
3. Evaluate the risks and decide whether the existing control measures are adequate or whether more should be done
4. Record the findings
5. Review the assessment and revise if necessary

In practice, most sectors of the aviation industry have already had some form of risk assessment applied to them and already have some form of control measures in place. In this situation the following steps may be useful:

1. Identify the tasks / areas to be assessed
2. Analyze and break down the tasks into manageable pieces
3. Identify and list all the hazards
4. Record and review the existing control measures (starting with the most effective)
5. Assess the remaining level of risk using the risk assessment matrix (see below)
6. Decide whether further action is needed to reduce the level of risk as low as reasonably practicable
7. Complete and communicate the assessment results
8. Review the assessment regularly (not less than once a year), whenever there is a change to The circumstances, or when there is an accident, incident or serious occurrence

The process of carrying out a risk assessment involves a trained risk assessor and a group of experienced individuals familiar with the task. The hazards need to be identified and the probability of them occurring should be tabulated in a risk assessment matrix (an example of a risk assessment matrix is shown following).

There are a number of matrices currently in use with varying definitions for the values for probability and consequence. The template shown above is used by many entities worldwide and is considered to be appropriate for the purposes of a risk assessment. The actual detail of the risk assessment matrix is not so important; however, the value of this type of assessment is to prioritize certain risks for remedial action or review.

Risk assessments should be reviewed on a regular basis to ensure they remain valid. They should also be reviewed after any accident, incident or serious occurrence and compared with the initial assessment. This is particularly important to ensure any lessons learned from an accident, an incident or serious occurrences are incorporated in the risk assessment. Sometimes these lessons learned may not have been thought of when action plans were originally written. Here are some suggested areas for assessment:

Staff Tasks:

1. Aircraft marshaling
2. Bird and wildlife control
3. Runway inspections
4. Airside driving
5. Runway change procedures
6. Use of airport cleaning vehicles
7. Spillage clean-up process
8. Use of air bridges
9. Activation of low-visibility procedures (LVP)
10. Removal of disabled aircraft actions

Business Risks:

1. Runway incursions
2. Adverse weather operations
3. Aircraft collision on the ground
4. FOD damage to aircraft
5. Aircraft fire Fighting
6. Major bird strike causing accident
7. ATC evacuation / loss of ATC service

Risk assessments should be appropriately documented to create an overall airport risk portfolio. Wherever possible, a follow-up and control process should be implemented allowing the safety supervisor to easily update and manage the airport risk portfolio. Furthermore, certain risks will also need to be reviewed in cooperation with third parties. The safety supervisor should ensure there are appropriate communication and cooperation methodologies in place with the third parties at the airport so that any risks identified are communicated in a timely and appropriate fashion.

Risk Probability		Catastrophic A	Hazardous B	Major C	Minor D	Negligible E
5	Frequent	5A	5B	5C	5D	5E
4	Occasional	4A	4B	4C	4D	4E
3	Remote	3A	3B	3C	3D	3E
2	Improbable	2A	2B	2C	2D	2E
1	Extremely Improbable	1A	1B	1C	1D	1E

Risk management	Assessment risk index	Suggested criteria
Intolerable region	5A, 5B, 5C, 4A, 4B, 3A	Unacceptable under the existing circumstances
Tolerable region	5D, 5E, 4C, 4D, 4E, 3B, 3C, 3D, 2A, 2B, 2C	Acceptable based on risk mitigation. It might require management decision
Acceptable region	3E, 2D, 2E, 1A, 1B ,1C, 1D, 1E	Acceptable

End

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